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Soft Matter in Lipid–Protein Interactions

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Abstract

Membrane lipids and cellular water (soft matter) are becoming increasingly recognized as key determinants of protein structure and function. Their influences can be ascribed to modulation of the bilayer properties or to specific binding and allosteric regulation of protein activity. In this review, we first consider hydrophobic matching of the intramembranous proteolipid boundary to explain the conformational changes and oligomeric states of proteins within the bilayer. Alternatively, membranes can be viewed as complex fluids, whose properties are linked to key biological functions. Critical behavior and nonideal mixing of the lipids have been proposed to explain how raft-like microstructures involving cholesterol affect membrane protein activity. Furthermore, the persistence length for lipid–protein interactions suggests the curvature force field of the membrane comes into play. A flexible surface model describes how curvature and hydrophobic forces lead to the emergence of new protein functional states within the membrane lipid bilayer.

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BACKGROUND AND SCOPE

A greater appreciation of membrane lipid–protein interactions (19) has the potential to significantly impact our understanding of biology at its confluence with physics and chemistry (3, 47, 76, 93, 104, 110, 129, 164, 165). Couplings of membrane proteins with water and the lipid bilayer (20, 48, 162) can profoundly shape the actions of G-protein–coupled receptors (GPCRs) (15, 41, 81, 88, 115, 153), ion channels (3, 127, 129), and transporters (12, 92). Membrane proteins are amphiphiles, which distinguishes them from the globular and fibrous proteins that are more thoroughly investigated at present. Crystal structures of membrane proteins (33, 36, 56, 78, 124, 135, 154) offer penetrating glimpses into their inner workings, yet they are static snapshots that do not correspond to the natural functional state in vivo (139, 155). With the abundance of new structures, until now, a static depiction does not fully explain how membrane proteins carry out their actions. Considering structure-function relations in the natural lipid bilayer requires approaches that are highly synergistic with X-ray crystallography of membrane proteins.

Biomembranes are supramolecular assemblies (3, 5, 11, 95, 98, 136, 148, 164), and thus we need to look beyond the crystalline state of proteins to grasp their roles at the molecular and cellular levels (19). Even for all-atom molecular dynamics (MD) simulations of membranes (80, 86, 151), current strategies tend to reduce atomistic detail to collective features of the liquid-crystalline molecules (77, 150). The van der Waals surfaces of proteins and other biomolecules are perhaps most readily visualized by their crystal structures. Yet other properties are also important (e.g., the spontaneous curvature or the bending elastic moduli that affect cellular membrane shape transitions). For a number of well-characterized proteins (12, 48, 52, 53, 93, 127, 129, 153, 158, 163) and peptides (3, 48, 69, 79, 143), structural and functional data point to modulation by the liquid-crystalline bilayer, including rhodopsin (13, 15, 153, 156), mechanosensitive and other channels (3, 127, 129), endophilins and BAR domains (146, 147), and transporters (12). Investigations of model lipid bilayers (84, 89), raft-like lipid mixtures (1, 151, 161), recombinant proteolipid membranes (15, 153), and simple natural membranes (97, 117, 133) all have contributed to our current understanding of their functions.

Previous accounts of lipid–protein interactions (10, 20, 93, 94, 129, 164) are highly informative, and together with detailed mathematical treatments (17, 72, 87, 108, 122) afford almost boundless inspiration. The busy reader may well be inclined to ask, what is unique about the present review? Here, our goal is to emphasize simple experimental and theoretical approaches that guide more extensive formulations. There is a large gap between the level of theory and the experimental investigations of the forces acting on the lipids and proteins. We naturally gravitate to the mesoscopic regime, falling between a molecular description (80, 86) and a continuum picture (9, 29, 65, 87, 108, 122, 129). Elucidating the role of the membrane lipids plus water—so-called soft matter—is a central theme (84, 150, 151). Concepts that have proven useful in explaining how lipid–protein (or peptide) interactions influence cellular functions are explored. We consider the various theoretical frameworks (with a minimum of mathematical detail), together with representative experimental data. The overall picture is that nonspecific biophysical properties of the lipids significantly affect protein-mediated functions of biomembranes, in which elastic deformation and softness of the bilayer play key roles.

G-protein–coupled receptor (GPCR): a protein that mediates signaling pathways involving hormones, neurotransmitters, vision, olfaction, or taste with a seven-transmembrane helical structure

Spontaneous curvature: the tendency of a lipid monolayer to curl as a result of an imbalance of attractive and repulsive forces involving the polar head groups and nonpolar chains

Rhodopsin: the G-protein–coupled receptor with 11-*cis*-retinal as its ligand that triggers the visual process upon light absorption

FUNCTIONAL LIPID–PROTEIN INTERACTIONS

Evidently, there are two lines of thinking that are prevalent with regard to biomembrane function. One school of thought is that the membrane lipids provide a neutral backdrop to the activities of membrane proteins and biologically active peptides (148). In this view, the lipid bilayer acts as a

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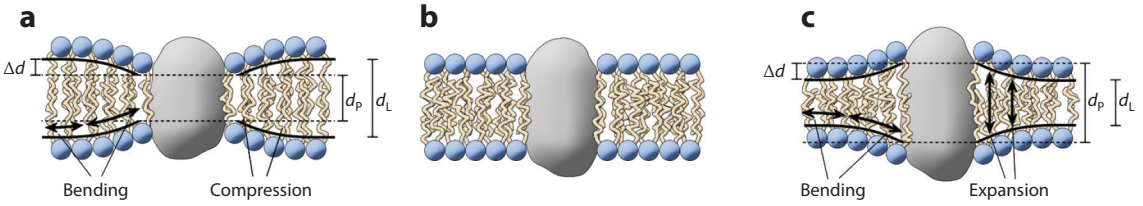

Figure 1

Integral membrane proteins are embedded in the lipid bilayer and solvated by the hydrocarbon chains of the lipid molecules. Proteins are depicted as solid-like objects; the lipid head groups are shown as circles and acyl chains as wormlike strings. Because of hydrophobic matching, the thickness of the hydrophobic proteolipid interface (d_P) is (*a*) less than, (*b*) equal to, or (*c*) greater than the mean bilayer thickness (d_L). Bilayer deformation entails bending plus compression or stretching of the hydrocarbon chains. Figure adapted with permission from Reference 22.

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permeability barrier to ions and polar molecules and enables membrane fluidity, while presenting different faces to the extracellular milieu and the cytoplasm (**Figure 1**). The standard fluid-mosaic model (FMM) encapsulates this thinking (148). It says that membranes are two-dimensional solutions of amphipathic proteins dissolved within a fluid lipid bilayer solvent (148) that facilitates their lateral and rotational diffusion, and gives the needed flexibility for protein conformational changes. Vectorial orientation of membrane proteins and lipids relative to the cytoplasmic and extracellular-facing monolayers (leaflets) enables processes such as signaling, protein folding, and trafficking of lipids and other biomolecules to occur.

Soft Matter and Membrane Function

The alternative view is that the lipids are more actively engaged with membrane protein-linked functions, which has gained currency recently (129, 136, 144, 165). Accordingly, membrane proteins are more strongly coupled to the lipids than previously considered, where the lipid properties are implicated in protein-mediated functions of biomembranes (52, 153, 163). Evidence has now accumulated that the lipid composition is among the thermodynamic state variables that govern protein function in membranes. Initial clues came from Ca^{2+}-ATPase (92) and rhodopsin (52, 113), together with studies of peptide ion channels (3, 79), amphipathic peptides (48, 69), and subsequently mechanosensitive ion channels (127, 129). Somewhat later on, the role of proteolipid coupling in biomembrane function received a boost from the raft hypothesis (16, 145). This picture has been recast by considering nonideal mixing of membrane lipids; or instead, critical behavior might play a role in solvating membrane proteins (67). The role of chemically nonspecific bilayer properties in modulating the functions of membrane proteins (19) is where the current picture introduces the new biophysics.

Fluid-mosaic model: the standard model in which biomembranes are viewed as two-dimensional solutions of randomly distributed proteins and lipids

As essential background, knowing about basic lipid biophysics and cholesterol interactions takes on great importance. A molecular-level understanding of the forces acting on lipids and proteins in membranes rests on knowledge of pure bilayer dynamics and thermodynamics in the absence of complicating protein molecules. For example, nuclear magnetic resonance (NMR) spectroscopy provides knowledge of average structure (142) and lipid mobility (27, 95), and has been applied to investigate the influences of ions (25, 141), water (84), cholesterol (1, 26, 106, 114, 161), and proteins (15, 152). Segmental order parameters of the lipids are directly accessible (95), as interpreted by Seelig (142) and others (44, 128) in terms of local bilayer structure and dynamics (27). Moreover, collective lipid interactions underlie the longer-range elastic properties of the membrane film (18, 95). Because of tethering of the lipids to the aqueous interface, collective

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FLUID-MOSAIC MODEL

The fluid-mosaic model (FMM) is currently the standard model for cellular membranes. It states that biomembranes constitute a two-dimensional solution of amphipathic proteins within a fluid lipid bilayer solvent. Integral membrane proteins are embedded in the lipid bilayer and solvated by the hydrocarbon chains of the lipid molecules. A fluid mosaic of oriented membrane lipids and proteins allows for the mobility needed for biological functions to occur that rely on molecular motions. The fluid-mosaic structure considers either weak coupling by nonspecific associations of lipids with integral membrane proteins or strong coupling due to chemically specific interactions with either the head groups or the acyl chains. According to the FMM, coupling interactions of the phospholipids and proteins of membranes are similar to the interactions among the lipids themselves and are relatively independent.

modes emerge on the mesoscale of the bilayer thickness and less as a result of the bulk membrane elasticity; yet the bilayer core resembles a simple hydrocarbon fluid (18, 95). Cholesterol yields increased bilayer stiffness and dampens the quasielastic membrane excitations at an atomistic level (106, 114), where its effects on lipid miscibility underlie nanoscale heterogeneities or inclusions with proteins known as rafts (1, 144, 161).

The Standard Fluid-Mosaic Model

By contrast, the FMM regards either weak coupling of lipids to integral membrane proteins or, alternatively, strong coupling involving either the polar head groups or the acyl chains (148). It asserts there is "no significant indication that the association of proteins with the phospholipids of intact membranes affects the phase transitions of the phospholipids themselves," suggesting "the phospholipids and proteins of membranes do not interact strongly; in fact, they appear to be largely independent" (148, p. 722). However, it is now understood that lipid–protein interactions do indeed affect the phase transitions of the phospholipids—but not the gel to liquid-crystalline transition (148) as originally thought. Rather, it is the transition from the lamellar phase to the reverse hexagonal (H_{II}) (or cubic) phase of the membrane phospholipids that tends to be affected by the proteolipid coupling (43, 131). As one example, the native retinal rod disk membranes contain bilayer-forming and nonbilayer-forming lipids (43), yet they are entirely in the fluid, liquid-crystalline (L_α) state (also known as liquid-disordered, or ld, phase) near physiological temperature (43, 111). The lipids are enriched in highly polyunsaturated omega-3 fatty acids, whereas cholesterol is present in minor amounts (20); lateral phase separation is absent, and lipid rafts are therefore questionable in this system (111). Evidently, the energy cost due to the proteolipid interface is compensated or balanced by the tendency of lipid monolayers to bend—there is a competition of opposing contributions, in which the membrane elasticity plays a key role (19, 21). (See the sidebar entitled Fluid-Mosaic Model.)

Cholesterol: a rigid molecule with a steroid ring structure and a simple hydroxyl group as the polar head found in the plasma membranes of animal cells

Rafts: small nanoscopic domains of cholesterol plus high-melting lipids and proteins that do not mix ideally with low-melting lipids

Lipid Influences on Protein-Mediated Functions of Biomembranes

In their role as modulators of protein activity, membrane lipids are becoming increasingly understood (12, 104). The influences of membrane lipids on protein functions are well described in the insightful reviews by Lee (92–94). Whether the lipids exert their influences because of molecularly specific interactions, or rather because of material properties of the bilayer, is one of the key questions addressed here (21, 76, 94, 104, 118). It is possible that the lipid influences correspond to direct binding to embedded membrane proteins; for example, specific lipids can act as allosteric

modulators, as in the case of GPCRs or ion channels (41). Alternatively, biophysical properties (19, 81, 115, 127, 129) can be involved. Renewed emphasis in this field is prompted by recent advances in solving the structures of a number of integral membrane proteins (56, 78). The new structural information adds significantly to our knowledge of the lipid influences on membrane protein function (94). Yet X-ray structural snapshots need to be combined with spectroscopic and functional studies for a full understanding to be achieved.

CHEMICAL SPECIFICITY OR BIOPHYSICAL PROPERTIES?

The proposal that chemically nonspecific properties of the bilayer directly affect the conformational energetics of membrane proteins (19) is based on experimental data first obtained for rhodopsin (43, 52, 163). Parallel studies of the influences of membrane lipids on the activity of Ca^{2+}-ATPase (92, 121) as well as investigations of nonlamellar-forming lipids in governing the activities of membrane-bound enzymes (7, 12, 39, 48, 74, 75, 121, 134) have also been important. Lipid influences attributable to bilayer deformation have been established for antimicrobial, ionophoric, cytotoxic, and fusion peptides (3, 48, 63, 69, 79, 96), as well as protein folding in membranes (38, 68, 116). Lastly, studies of the growth of microorganisms, as pioneered by Lindblom et al. (97), have uncovered a connection between lipid polymorphism and membrane function (97). How can we begin to unify these diverse biochemical and biophysical findings at the membrane level?

Beyond the Fluid-Mosaic Model

Evidence has now uncovered the striking influences of the membrane lipid bilayer on the activities of various membrane proteins (93), in which the tightly regulated lipid composition affects the protein function (8, 19, 53, 59, 74, 75, 93, 97, 113, 121, 163, 165). A balance of lamellar- and nonlamellar-forming lipids in *Acholeplasma laidlawii* (97) as well as *Escherichia Coli* (37, 46, 117, 132) is important for cellular growth, thus establishing the concept of homeostasis of spontaneous curvature as a determinant of membrane function. For rhodopsin, its signaling function involves the pH-dependent metarhodopsin I (MI) to metarhodopsin II (MII) equilibrium as a result of light activation. A direct effect of membrane lipids on the conformational energetics of an integral membrane protein was discovered for the first time (8, 19, 163) (**Figure 2**). Rather than a two-state switch, the endpoint for the equilibrium does not reach zero at higher temperatures, indicating the active MII form is an entropy-driven ensemble of states (100) (**Figure 2*a***). Moreover, a combination of the native head-group and acyl-chain composition yields native-like activation of rhodopsin by light, yet neither alone is sufficient (**Figure 2*b***). These findings have been seminal for current thinking regarding the roles played by the lipids in cellular membranes (30, 32, 59, 102) and suggest the standard FMM is overdue for revision.

For rhodopsin, the influences of both the membrane lipids and water (35) are explained by an ensemble-activation mechanism (155), where additional lipid substitution experiments clearly show the following: (*a*) The membrane lipid composition can significantly perturb the formation of active MII rhodopsin; (*b*) the perturbation of the metarhodopsin equilibrium is thermodynamically reversible and can be driven forward or backward depending on the state variables (T, P, lipid to protein ratio, osmotic stress); (*c*) changing the lipid head groups can compensate for changing the acyl chains, indicating lack of chemical specificity; (*d*) proximity to a lamellar–nonlamellar lipid phase boundary promotes the rhodopsin activation; and (*e*) the metarhodopsin equilibrium depends on the lipid to rhodopsin ratio. Chemically nonspecific bilayer properties are likely as determinants of rhodopsin function (15), because changing the lipid head groups can substitute for

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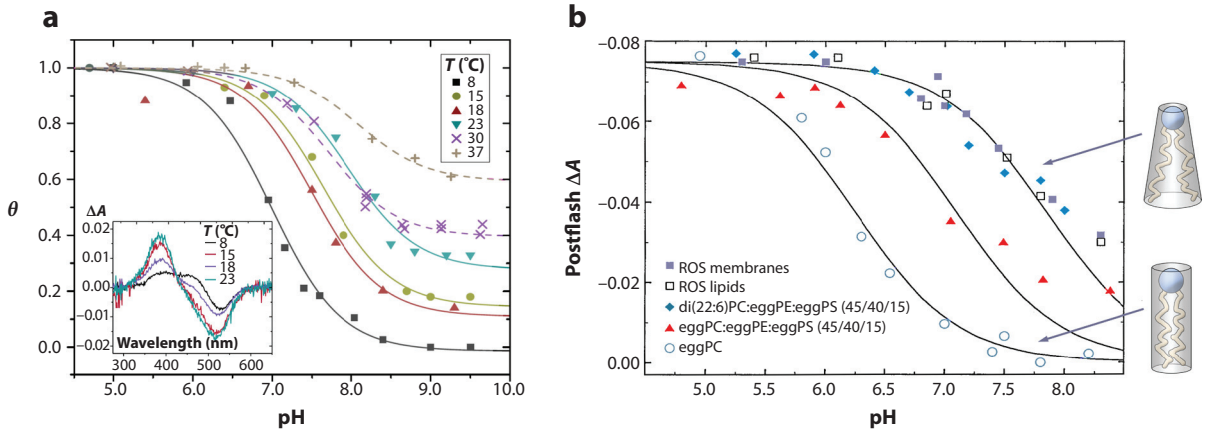

Figure 2

Light activation of rhodopsin is due to an ensemble of states involving the membrane lipids. (*a*) Fraction of active metarhodopsin II (MII) state (θ) in equilibrium with inactive metarhodopsin I (MI) state as a function of pH. At higher temperatures, the alkaline endpoint does not reach zero because of the active MII substates. (*Inset*) Ultraviolet-visible difference spectra (light minus dark) reveal how the MI–MII equilibrium depends on temperature. (*b*) Postflash absorbance change ($\lambda = 478$ nm because of active MII) is plotted versus pH at $T = 28°C$. Note that both the apparent pK_A and the alkaline endpoints depend on the head-group and acyl-chain composition of the membrane phospholipids (indicated in figure). Figure modified with permission from Reference 20. Abbreviations: PC, phosphatidylcholine; PE, phosphatidylethanolamine; PS, phosphatidylserine; ROS, rod outer segment.

changes in the acyl chains, and there is an influence of the lipid to protein molar ratio. Attractive and repulsive forces corresponding to the average lipid shapes and/or membrane curvature forces most likely play a role (**Figure 2*b***).

Short-Range Lipid–Protein Interactions: Annular or Boundary Lipids

In a protein-centered universe, there are significant possibilities of functional lipid interactions. Spin-label electron paramagnetic resonance (EPR) spectroscopy has been important for revealing how short-range solvation of integral membrane proteins involves a single shell of annular or boundary lipids (57, 104). Recently, attention has been drawn to lipid structural specificity (93, 94) by the X-ray crystal structures of integral membrane proteins, together with applications of native mass spectrometry (90, 107). The surfaces of membrane proteins contain many shallow grooves or protrusions, to which the fatty acyl groups of the surrounding lipids can conform, allowing proteins to be tightly packed within the membrane lipid bilayer. The lipid molecules resolved in X-ray crystal structures of membrane proteins tend to be found in clefts between the transmembrane (TM) helices, or to occupy a partial belt about the protein hydrophobic surface (94). Phospholipids and cholesterol can occupy the space between the TM helices, corresponding to a somewhat lower specificity than for allosteric binding to specific protein sites. Molecular dyamics simulations also support this location, where lipid acyl chains such as docosahexaenoic acid are found to localize between the TM helices of rhodopsin (58). A relatively nonspecific structural role due to lipid solvation is suggested by the belts of lipids surrounding integral membrane proteins in X-ray crystal structures (94), and by the annular (boundary) lipids detected using native mass spectrometry (107). Effectively, the X-ray, spin-label EPR, and native mass spectrometry studies converge to show how the flexible lipids act as a solvent for integral membrane proteins. To ensure membrane proteins are compatible with the lipid bilayer structure, their insertion must

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not make the membrane leaky, which would compromise the essential permeability barrier upon which all life depends.

Solvation of the Proteolipid Interface

However, the new twist for membrane proteins is that there are two faces—one needs to consider solvation of both the aqueous regions and solvation of the nonpolar regions (**Figure 1**). For instance, perturbation of water is found in X-ray crystal structures to extend mainly to the first solvation shell, and not beyond the second shell of water about the protein. According to spin-label EPR spectroscopy (57, 104), the perturbation of the lipids likewise includes mostly the first solvation shell. Here, one should recall that the hydrophobic effect is short range in nature, meaning that one mainly considers the molecularly thin interface between hydrocarbon and water (21, 45). Structures of the lipids adjacent to the protein can be distorted, because of the protrusions and grooves or clefts between the TM helices, yet they are swapped rapidly with the bulk lipids by lateral diffusive exchange (23). There is generally only low selectivity for the head-group composition of the annular lipids; hence, the composition of the boundary (annular) lipids resembles the bulk lipid bilayer (93). As one example, rhodopsin has marginal selectivity for different polar head groups, and additional proteins typically show a low selectivity for anionic phospholipids (93).

Even so, for various lipids at the bilayer aqueous interface, differences in hydrogen bonding and hydration can occur, as in the case of phosphatidylcholine (PC) versus phosphatidylethanolamine (PE) (128). Comparing PC to PE, there are clearly variations in head-group size, hydrogen bonding, and hydration. The hydrophobic $-N(CH_3)_3^+$ group of PC induces clathrate-like structures about the head groups to enable hydrogen bonding to water molecules; whereas for PE, direct hydrogen bonds between the $-NH_3^+$ groups and water molecules condense the area per lipid head group and yield reduced hydration. Because the area per lipid is inversely related to the bilayer thickness (128), formation of the proteolipid interface involves matching the hydrophobic thickness of the membrane, possibly together with curvature deformation (see below).

Long-Range Lipid–Protein Interactions: Emergent Properties

Notably the influences of the protein solvation by the lipids and by water are expected to extend only to a single shell of molecules surrounding the protein groups (57, 104). A striking counterexample, first reported by Botelho et al. (14), is the influence of the lipid to protein molar ratio on the light activation of rhodopsin: Increasing the number of lipids yields greater stabilization of the active MII state (15). One explanation entails crowding of the protein molecules within the bilayer, as opposed to lipid–protein interactions. But why invoke crowding when lipids far away from the protein, beyond a single annulus, obviously can affect its conformational states? Evidently, the length scale of the molecular forces is large compared to the molecular size—a fundamental caveat of elasticity theory. Though it costs energy for lipid acyl chains to solvate the apolar protein interface, long-range interactions of the molecules can also come into play (19, 21, 163).

The question of long-range lipid–protein interaction is where the new biophysics is expected to lie. Our application of physics-based approaches to lipid–protein interactions differs fundamentally from the standard FMM (53, 163). In a lipid-centered universe, the emphasis is put on the lipid material properties, such as the work needed for deforming the protein shape within the bilayer (19, 20, 138). We now ask, how short are the distances we should consider before atomic- or molecular-size effects (98, 125) become significant? That brings us to the topic of the actual models that have been introduced for the lipid–protein interactions at both the atomistic and the mesoscopic levels.

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Below, we critically consider each of these proposals by evaluating its characteristic strengths and weaknesses.

Hydrophobic matching: the principle that the hydrophobic protein surface matches the hydrocarbon thickness of the lipid bilayer

HYDROPHOBIC MATCHING AND BILAYER THICKNESS

Perhaps hydrophobic matching is the most obvious property that might account for nonspecific lipid–protein interactions. In accord with the FMM, the lipids act as a two-dimensional solvent for the embedded membrane proteins, in which hydrophobic matching to the intramembranous proteolipid boundary is connected with the lateral pressure profile. Structural correlates come from spin-label EPR spectroscopy (57, 104) and the observation of lipid belts surrounding integral membrane proteins in X-ray structures (54). Differences in activity for various membrane proteins due to changing the fatty acyl length (8, 15, 20, 92–94) indicate that membrane proteins can distort or change their conformation owing to hydrophobic matching. Experimental support comes from studies of the effects of the lipid acyl length on protein activity for Ca^{2+}-ATPase (92) and rhodopsin (19, 20, 153). In the case of rhodopsin (19, 20, 152), the MI–MII activation equilibrium depends on the lipid acyl-chain length, where a thicker bilayer (128) favors the active MII state (see **Figure 1**). That leads us to the idea that the equilibrium can be tipped toward one state or the other, depending on structural features such as the bilayer thickness matching. Additional head-group interactions may be important, which in the case of PE can be due to hydrogen bonding, size, or hydration; nonspecific electrostatics for phosphatidylserine (PS) (52) and phosphatidylglycerol (PG); or specific charge interactions for cationic lipids (34).

Membrane Deformation and the Proteolipid Interface

The principle of hydrophobic matching (3, 49, 76, 92, 94, 104, 118, 119) states that the physical constraints imposed upon integral membrane proteins involve a fluid mechanical coupling of their nonpolar surface to the bilayer thickness (85). Models of hydrophobic matching (76, 119) typically consider that the lipid acyl chains surrounding membrane proteins adapt their projected length to conform to its intramembranous surface and that the protein acts like a rigid body. A schematic depiction (**Figure 1**) illustrates how hydrophobic mismatch could yield distortion of a lipid bilayer around a membrane protein. Examples are shown for a protein whose hydrophobic thickness is less than the lipid bilayer (*left*; $d_P < d_L$), equal to (*middle*; $d_P = d_L$), or greater than the lipid bilayer (*right*; $d_P > d_L$). Stretching or compressing the lipid chains gives a decrease or increase in the cross-sectional area per lipid (i.e., the area per lipid at the aqueous interface). If the hydrophobic thickness of the protein exceeds the bilayer thickness, the lipid chains must be stretched to solvate the protein; hence, the area per lipid molecule adjacent to the protein is less than for bulk lipids. Conversely, hydrophobic matching of a protein to a thick membrane compresses the fatty acyl chains of lipids, so there is a greater area per lipid at the aqueous interface. The free energy cost of the hydrophobic mismatch goes as $|d_P - d_L|^2$ in a harmonic approximation, where d_P denotes the intramembranous protein distance and d_L is the lipid hydrophobic thickness (119) (see **Figure 1**). Alternatively, the influence of the bilayer thickness can also be formulated in terms of the mean area per lipid at the aqueous interface, including how it is affected by interactions with the embedded membrane proteins (84, 128).

The Elusive Grasp of Membrane Lipids

For integral membrane proteins, solvation of their hydrophobic surface is related to the (tensile) stress as a result of stretching the lipid chains or the compressive stress (pressure). Possible effects of

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hydrophobic mismatch are lipid chain ordering, phase behavior, and microdomain (raft) formation (82). Broadly considered, hydrophobic mismatch can entail either adaptations in helix tilt or rotations of the amino acid side chains (such as tryptophan) at the ends of the TM helices. Long helices are most likely able to match thin bilayers by a change in tilt and/or rotations of the helix, together with rotation of the amino acid residues at the ends of the helices. Support for changes in helix tilt comes from solid-state NMR studies of membrane-bound peptides (126, 159). However, the molecularly rough surface of a membrane protein is in contact with the surrounding bilayer and can result in poor packing with the lipids, unless they distort to match its surface. Tilting of the lipid chains and conformational disorder can lead to optimal sealing of the protein into the bilayer, thus allowing the membrane to act as a permeability barrier. Consequently, we expect the intramembranous hydrophobic surface of a protein to be solvated by a shell of disordered lipids, in analogy with the role of water in solvating globular proteins in solution.

Persistence length: a mechanical property that quantifies the stiffness of a polymer or a flexible surface

Although hydrophobic matching is simple and intuitively appealing—and persuasive at first glance—it focuses mainly on the short-range hydrophobic effect. Further consideration indicates that it may overlook collective interactions of both the lipids (18, 28) and proteins (15, 21). Among the open questions are the properties and role of the boundary lipids in the first solvation shell about membrane proteins; whether hydrophobic mismatch yields distortion of either the lipid bilayer or the embedded proteins or peptides; and the role of the lipid phase state (e.g., gel, liquid-crystalline, reverse hexagonal) in the proteolipid coupling. Experimentally, solid-state ^{2}H NMR spectroscopy shows that under matched conditions, the distortion of the lipid bilayer about integral membrane proteins like rhodopsin tends to be relatively small (70, 152). Apparently, the energetics of the deformation are rather high, and small structural changes can have a large energetic penalty. If bilayer distortion does not yield complete hydrophobic matching, then the protein itself could distort (by a conformational change) to match the hydrophobic thickness. There could be a restructuring or change in the TM helix tilt, or residues (such as tryptophan) at the ends of the helices could rotate or change their position versus the aqueous interface.

Short-Range Solvation Versus Long-Range Proteolipid Couplings

Still, does short-range solvation provide the full answer to explaining functional protein conformational changes? The fundamental difference between membrane proteins and globular proteins is that the forces acting upon the latter are approximately isotropic over the surface. By contrast, the forces are significantly anisotropic for the case of membrane proteins, meaning they exist in the curvature stress field of the lipid bilayer (19). The forces are not averaged over the protein surface, and thus long-range elastic interactions can come into play. Because the hydrophobic effect is short range, there should not be an influence of the lipid to protein ratio on membrane peptides and proteins. Yet this is contrary to experimental tests, which show an increase in the fraction of light-activated rhodopsin (MII state) (15, 123, 152). For now, we note that the influence of the lipid to protein ratio implies there is a persistence length for the perturbation upon moving away from the proteolipid boundary. The idea of a persistence length is consistent with a role of curvature forces in the elastic membrane deformation (64, 65).

LIPID MIXING AND RAFTS IN CELLULAR MEMBRANES

But initially let us turn to the topic of lipid rafts in mammalian cellular membranes. The subject has been insightfully discussed (1, 16, 42, 144, 145), and only a brief summary will be given here. Cellular membranes are very heterogeneous in their lipid and protein compositions, and raft-like microdomains or clusters are proposed to occur in the 10–100-nm size range, owing

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to collective interactions among the molecules (1, 144). In mammalian cells, typically rafts are associated with lipid mixtures enriched in cholesterol and high-melting lipids like sphingomyelin, which can separate from lower-melting lipids in the bilayer (55, 161). They can affect protein sequestration and signaling, membrane trafficking, virus budding, and other cellular processes. Models for raft formation often invoke lipid mixtures containing cholesterol that phase-separate into liquid-ordered (lo) and liquid-disordered (ld) regions (161). When cholesterol and high-melting lipids like sphingomyelin or PE are present in mixtures with lower-melting lipids like PC, nanoscale heterogeneities or phase separations of lo regions from ld regions can occur (1, 16, 55). Broadening of the order–disorder transition due to cholesterol could allow additional interactions of lo or ld regions with proteins, which could affect their activities (1, 144).

Miscibility critical point: the conditions where a lipid mixture has a fluctuating local composition with transient domains enriched in one of the lipid species

With regard to raft-like lipids the lipid–protein interactions necessarily involve cholesterol. Because of nonideal mixing, the influences of proteins on raft-like lipid mixtures can differ from their effects on the main order–disorder [solid-ordered (so)–ld] phase transition of the lipids. For the raft-like (lo) nano- or microstructures, the lipid–protein interactions are analogous to the lipid interactions (145). The idea of weak lipid–protein couplings is consistent with the assumptions of the FMM as described above. The lipid–protein coupling is comparable to interactions of the lipids among themselves, either in the ld state, in lipid rafts, or in the case of liquid–liquid immiscibility. For weak lipid–protein couplings, essentially the lipid properties affect the protein oligomeric states or lateral distribution (e.g., in the case of raft-like microdomains). The lipids would push the proteins around. There could be a two-way coupling or balance, in which the system could be tilted from one state to another, depending on interactions of the components.

When Cholesterol Is Lacking: Lamellar to Nonlamellar Phase Transitions

At this point, we can conclude that both the order–disorder phase transitions and the lipid rafts appear only weakly affected by the proteolipid coupling. However, this proposal does not consider the lamellar to nonlamellar phase transitions of lipids with fluid, liquid-crystalline chains (see below). Lamellar to nonlamellar phase transitions can be more strongly affected, owing to the presence of proteins, which in the case of rhodopsin is found to stabilize the lamellar phase (43). The strong proteolipid coupling represents a significant departure from the FMM (148). One can expect that strong curvature forces give rise to the lamellar to nonlamellar phase transitions, and might also be coupled to the protein stability (21). Even so, let us first take into account whether the physical state of the lipids may be analogous to critical fluids, involving proximity to a miscibility critical point in the temperature-composition phase diagram (67).

BIOMEMBRANES AS CRITICAL SYSTEMS

It has also been proposed that lipid membranes may be analogous to complex two-dimensional fluids, whose properties may be closely linked to protein-mediated biological functions. Lateral phase separation and critical behavior have long been under consideration with regard to collective lipid–lipid and lipid–protein interactions (109, 118). For rafts (1, 144, 161), the above ideas have been recast, such that the cellular membrane lipid composition is close to a miscibility critical point (66). Critical systems have a rather large susceptibility, meaning they have large responses to small inputs—the system can be tilted from one state to another with greater sensitivity than would be possible otherwise. As one example, carbon dioxide has a critical point near atmospheric pressure and temperature and is used as a supercritical solvent for decaffeinating coffee. In biology, large-scale systems can have highly collective behavior triggered by small inputs from external forces, as in the schooling behavior of fish or in the neural networks in the retina.

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Homeostasis of Criticality

Earlier proposals have suggested the lipid composition is naturally selected or controlled, so that the growth temperature is at an optimum relative to the order–disorder phase transition (37). In the case of biomembranes, the collective interactions encompass clusters of the proteins and lipid molecules, where structural variables of the proteolipid assembly include the bilayer thickness or area per lipid molecule (84, 128). Accordingly, the lipid compositions of plasma or organelle membranes of animal cells can be selected or tuned to be close to a critical point under the growth conditions (e.g., as revealed by fluorescence microscopy) (66, 67, 160). In the vicinity of a miscibility critical point, nanoscale collective interactions among the lipid or protein components enable large-scale fluctuations of the system that involve a two-way coupling of the dynamics. Homoeostasis of criticality could be applicable to cellular plasma membranes with high levels of cholesterol. Extension of results for model lipid mixtures to complex biological membranes has the potential to elucidate novel insights (160, 161). For example, living cells could adjust their proximity to a miscibility critical point upon changes in growth temperature, changes in nutrient conditions, or the presence of general anesthetics. Small perturbations of the lipids may thus tilt the system from one protein functional state to another.

Critical Fluctuations in Cellular Membranes

Interactions between membrane proteins could thus be modulated by sensitive tilting points, which trigger various biochemical signaling pathways or networks. For model lipid mixtures containing cholesterol, at subcritical or near-critical temperatures, collective fluctuations are observed by fluorescence microscopy (66) and NMR spectroscopy (161), with correlation lengths in the 10-nm up to the micrometer range. At the miscibility critical point of a lipid mixture, the tie lines of the phase diagram merge to single points. The compositions of the coexisting phases become equal, and large-scale fluctuations occur, as in the case of critical opalescence. Ternary lipid mixtures with a critical composition traverse a miscibility critical point with changing temperature (161), where the correlation length for the critical compositional fluctuations diverges as the miscibility critical point is approached (67, 160). Besides ternary lipid mixtures, critical behavior is proposed for the plasma membranes of living cells (160). Upon cooling, plasma membrane vesicles phase-separate into coexisting liquid-like regions, as observed using fluorescence microscopy (160). The correlation length depends on the proximity to the critical point in a temperature-composition phase diagram, meaning that a large range of sizes of the putative raft-like clusters could occur in cellular membranes (10-nm to the micrometer range). Mammalian cell plasma membranes may be tuned to criticality to sense and respond to environmental stimuli.

Theoretical Models for Critical Fluctuations

In animal cells, the membrane composition can be naturally selected or predisposed to criticality; for example, near the miscibility critical point, micron-sized fluctuations are evident with fluorescence microscopy (160), which obey universal scaling laws. Such membranes exhibit critical phenomena of the two-dimensional (2-D) Ising universality class, meaning the critical behavior is independent of the interactions that give rise to the critical point (66, 67). The case of a 2-D Ising model predicts the correlation length of the lipid fluctuations varies inversely with temperature away from the critical point (160). For subcritical or near-critical fluctuations, the submicrometer fluctuations occur down to some tens of nanometers. A broad distribution ranging from the segmental fluctuations of the individual lipids up to collective fluctuations of the entire bilayer is

SPONTANEOUS CURVATURE

The spontaneous curvature of a membrane lipid film describes the balance of attractive and repulsive forces that underlies the lateral pressure profile. When the optimal separation of the polar head groups exceeds the chains, as for single-chain amphiphiles—including detergents or lysolipids, or double-chain gangliosides with large head groups—there is a positive spontaneous curvature (concave toward hydrocarbon). Small micelles or normal (type-1) hexagonal phases are formed. At high surfactant concentration, micelles are forced to pack into a cubic lattice with long-range translational order (micellar cubic phase). When the optimal head-group separation is the same as the chains, the spontaneous curvature is zero, as it is for saturated phosphatidylcholines. The planar bilayer is stabilized as in standard textbook representations. Lastly, when the optimal head-group separation is less than the chains, the spontaneous curvature is negative (concave toward water), as in saturated or polyunsaturated phosphatidylethanolamines. Reverse hexagonal (type-2) phases or bicontinuous cubic phases are formed. In the latter case, there are two noninterpenetrating aqueous regions separated by a lipid film draped onto a lattice with cubic symmetry (plumber's nightmare).

shown by nuclear spin relaxation studies (24, 95). One consequence is that long-range entropic forces known as Casimir forces (99) may act upon the membrane-bound protein inclusions. Over 5–10-nm length scales and greater, the critical Casimir forces could affect or mediate long-range interactions among the protein molecules, such as those associated with raft-like heterogeneity in cellular membranes. Yet there are counterexamples of membranes lacking cholesterol (97), or low in cholesterol and sphingomyelin (20), where significant influences of lipid–protein interactions also appear to be operative. Plant cell membranes are very low in cholesterol; microbes are lacking cholesterol; and proteolipid systems exist that are in the ld (liquid-crystalline) state, with little evidence of criticality or liquid–liquid phase separations. Though fascinating, the proposal of collective fluctuations near a miscibility critical point may be restricted in its generality.

BIOMEMBRANES AS TRANSDUCERS OF CURVATURE STRESS

An alternative to considering lipid miscibility or chemically specific lipid–protein interactions involves a continuous membrane lipid film (3, 17, 19, 77, 129). Accordingly, the differential geometry of curves and surfaces (72) can be applied to study the membrane shape fluctuations or the deformations arising from the proteolipid couplings. Here, the idea is that the energetics of membrane curvature—the so-called language of shape (72)—play a role in governing the lipid–protein couplings (19, 53, 68, 69, 76, 82, 96, 119, 122). The free energy of elastic membrane curvature deformation affects lipid–protein interactions that modulate key biological functions, such as transport, ion conduction, and signaling. Protein conformational changes and stability (3, 19, 127, 129), folding (116), and membrane fusion (143) can all involve elastic deformation of the lipid bilayer. (See the sidebar entitled Spontaneous Curvature.)

Homeostasis of Curvature Elasticity

The option to regarding only hydrophobic mismatch or lipid criticality (see above) has been long under discussion (43, 163). It is known that both plant and animal membranes contain lipids with a tendency to form nonlamellar phases. Besides liquid–liquid immiscibility due to ordered or disordered lipids, curvature elasticity of the liquid-crystalline lipids can come into play. The hypothesis of homeostasis of spontaneous curvature was put forth in the 1980s for

Differential geometry: the branch of mathematics that deals with curves and surfaces

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integral membrane proteins (20, 21, 163) and for microbial growth (59, 97). It currently remains as an alternative to the FMM, and in the case of membrane proteins (13, 19), it is called the flexible surface model (FSM). Changes in protein shape involve the work due to integrating the molecular displacements versus the anisotropic force field (138). Here, one should recall that proteolipid membranes are different from the case of globular proteins in solution, where the forces are more isotropic. In this way, homeostasis of membrane spontaneous curvature explains the tightly regulated membrane lipid compositions of biomembranes through lipid–protein interactions (15).

Flexible surface model: the new membrane model in which the curvature stress field of the lipid bilayer governs the protein functional states

Balance of Forces and Molecular Packing

As one illustration, consider a schematic temperature-composition phase diagram for a hypothetical phospholipid (**Figure 3*a***). Such a phase diagram can be interpreted by various cuts through the continuous mathematical surface. For a given composition, the series of microstructures shows greater tendency to curl toward water as temperature rises. The sequence of structures with increasing temperature manifests the repulsive pressure of the hydrophobic acyl chains versus the polar head groups. Transitions from the planar lamellar phase (L_β or L_α) to the bicontinuous lipidic cubic phase (LCP) ($Im\bar{3}m$ or $Pn\bar{3}m$) and reverse hexagonal (H_{II}) phase reveal a more negative spontaneous curvature. Conversely, for a given temperature, increasing the water content results in the lipid polar head groups becoming progressively more hydrated. A sequence is found going from the reverse hexagonal (H_{II}) and LCPs to the lamellar phase, because of a less negative spontaneous curvature. The membrane lipids include a balance of forces that give curvature-dependent phase transitions between various nanostructures. [Note that the distinction in terms of solid-ordered (so), liquid-disordered (ld), and liquid-ordered (lo) phases (rafts) is inapplicable to membrane curvature.]

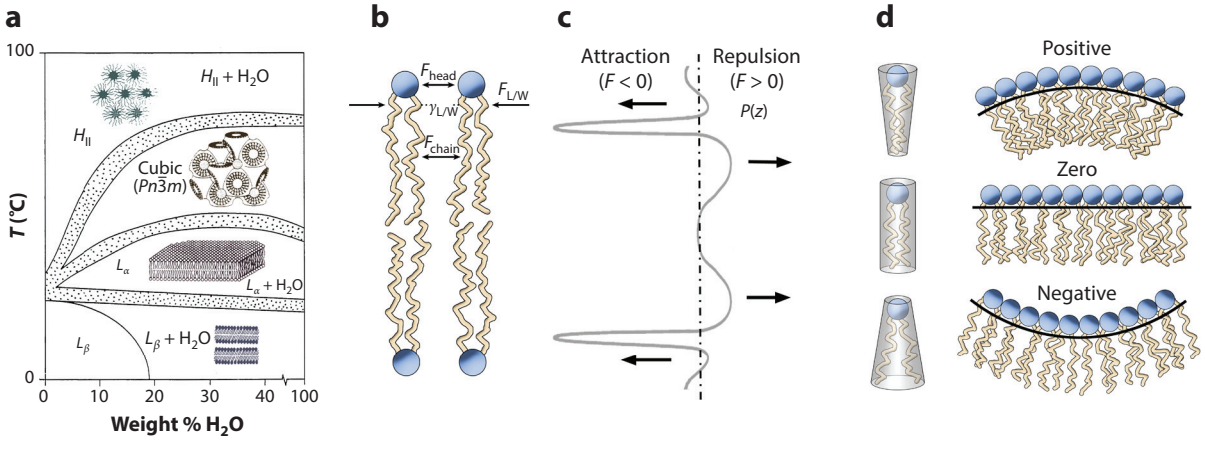

Figure 3

Phospholipids assemble into nanostructures as a result of a balance of forces among the polar head groups and nonpolar hydrocarbon chains. (*a*) Schematic phase diagram showing microstructures with increasing tendency to curve toward water as temperature rises. (*b*) Attractive force (pressure) acting at the lipid/water interface ($F_{L/W}$) due to the hydrophobic effect, which balances the repulsive force (pressure) of the head groups (F_{head}) and acyl chains (F_{chain}). (*c*) Corresponding profile of lateral pressure as a function of depth along the bilayer normal. (*d*) Illustration of molecular packing showing lipids with different head groups and acyl chains inscribed within their geometrical shapes. The spontaneous (intrinsic) curvature for a lipid monolayer is due to an imbalance of lateral forces: positive (toward hydrocarbon), zero (planar), or negative (toward water). Figure modified with permission from Reference 21. Copyright (2012) American Chemical Society.

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Lateral Pressure Profile

Next we can ask, why deal with an explicit proteolipid membrane (58, 70) if we can bypass an atomistic or molecular perspective? As described above, membranes entail a balance of opposing attractive and repulsive forces that underlies the structural polymorphism. **Figure 3*b*** depicts the attractive and repulsive forces acting at the level of the lipid polar head groups and the nonpolar acyl chains. Notably, the lateral pressure profile has three contributions (30, 32, 102, 103, 140). Within the head-group region, attractive and repulsive interactions at the polar–nonpolar interface govern the area per molecule (120, 128). (*a*) The attractive pressure ($F_{L/W}$) (negative) is due to the surface tension ($\gamma_{L/W}$) of the hydrophobic acyl groups with water (140), which tends to minimize the interfacial area. Additional attractive interactions include head-group dipole and hydrogen-bonding forces that act in concert with the long-range van der Waals force among the acyl chains of the two monolayers (128). (*b*) The opposing repulsive pressure (positive) is due to short-range steric forces, arising from both the head groups (F_{head}) and the acyl chains (F_{chain}). In the head-group region, the lateral pressure (F_{head}) arises from steric, hydration, and electrostatic effects. Though typically repulsive, it may contain attractive contributions from hydrogen-bonding interactions. (*c*) Lastly, within the chain region, the repulsive lateral pressure (F_{chain}) is due to thermally activated bond rotational isomerizations. Together, the repulsive head-group pressure and the acyl-chain pressure counterbalance the attractive lipid/water interfacial tension—the resultant lateral pressure is zero at equilibrium. The balance of attractive and repulsive forces governs the resulting lateral pressure profile of the lipids (**Figure 3*c***) and can affect their remodeling due to membrane biogenesis, interactions with proteins or peptides, or environmental signals.

The idea that a balance of opposing forces is manifested in the geometrical lipid shapes is embedded in an earlier explanation for the polymorphism of membrane lipids (140), due to Israelachvili and Ninham (73). For a given head-group size, the area per lipid constrains the packing of the acyl chains (128), thus yielding the observed microstructures. Such a viewpoint exemplifies the chemistry perspective (157), whereby molecular packing is quantitatively related to curvature (101) (**Figure 3*d***). The packing of lipids within the aggregate manifests attractive and repulsive forces acting upon the polar head groups and the nonpolar acyl chains (73). Amphiphiles with a greater head-group size relative to the chains, such as lysophospholipids, single-chain detergents, or gangliosides, favor packing into a conical molecular shape on average [**Figure 3*d*** (*top*)]. They form micelles or normal H_I phases and are analogous to an oil-in-water dispersion (140). Lipids with larger head groups—for instance, PC whose head group is methylated—tend to pack on average with a cylindrical molecular shape [**Figure 3*d*** (*middle*)]. They form a planar lipid bilayer as found in standard biochemistry texts. Last, those lipids with relatively small head groups compared to the chains, such as PE, pack into an inverted conical molecular shape on average [**Figure 3*d*** (*bottom*)]. They form the reverse H_{II} phase, which is analogous to a water-in-oil dispersion (140, 157). The idea of a molecular packing parameter corresponds to the lateral pressure profile, in terms of a balance of opposing forces acting at the level of the lipid polar head groups and the nonpolar acyl chains. Optimal packing of lipids in membranes is also directly related to the curvature energy (97), as discussed below.

The Monolayer Spontaneous Curvature

We now return to the lateral pressure profile, and one should note that it is invisible—it cannot be measured. Only theoretical simulations can determine the lateral pressure profile (29), as originally shown by Schulten and colleagues (61). The lateral pressures and tensions are distributed across an individual lipid monolayer (leaflet). Even so, the spontaneous curvature H_0 can be experimentally determined for membrane lipids under dual solvent stress (130). Mismatch of the optimal areas

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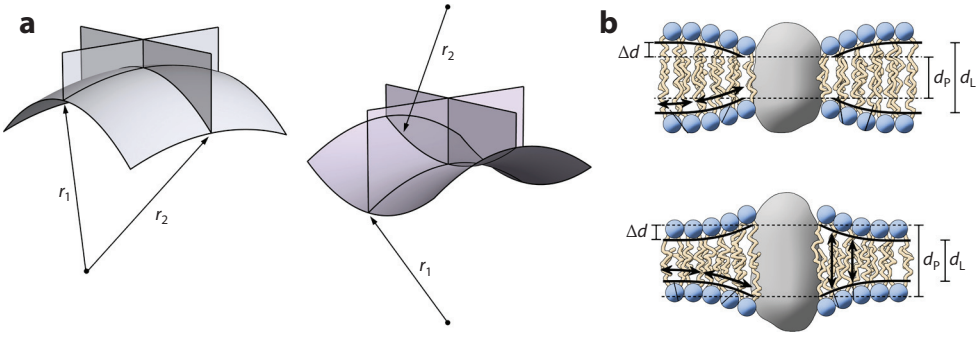

Figure 4

Language of shape explains how membrane lipids can influence protein function. (*a*) Membrane deformation is described by two principal curvatures. Note that the principal curvatures ($c_1 = 1/r_1$ and $c_2 = 1/r_2$) are of the same sign or opposite sign as indicated. (Figure courtesy of J. Kinnun.) (*b*) Flexible surface model (FSM) describes lipid–protein interactions by a new paradigm of shape. (*Top*) Hydrophobic thickness of protein (d_P) is less than thickness of the lipid bilayer (d_L), yielding compression of acyl chains. (*Bottom*) Hydrophobic thickness exceeds the bilayer thickness, giving stretching (expansion) of lipid chains. Competition of membrane bending with solvation of the proteolipid interface yields the functional protein states. Figure modified with permission from Reference 22.

of the head groups versus the cross-sectional chain area gives a bending moment for an individual lipid monolayer (101). Examples are shown (**Figure 3*d***) where the spontaneous (intrinsic) monolayer curvature (H_0) is positive (toward hydrocarbon), zero, or negative (toward water). The spontaneous monolayer curvature becomes more negative as temperature increases or hydration is less (greater tendency to curl), giving the observed sequence of nanostructures. Additional phases with significant curvature can occur, including microemulsions and bicontinuous cubic phases. The various nanostructures are characterized by their principal curvatures (**Figure 4*a***). They include the gyroid (G), Schwartz diamond (D), and primitive (P) minimal surfaces (where the mean curvature is everywhere zero) (e.g., the so-called plumber's nightmare); they are related through the Bonnet transformation (72). The lipid or surfactant mono- or bilayer is draped upon an infinite periodic minimal surface, giving a labyrinth-like system of channels.

Principal curvatures: the maximum and minimum curvatures of a regular surface

Mean curvature: the geometric average of the two principal curvatures of a regular surface

Minimal surface: a regular surface that locally minimizes its area and for which the mean curvature is zero

Plumber's nightmare: a bicontinuous cubic phase having interconnected aqueous channels with no pathway from one side of the dividing surface to the other

When the optimal head-group separation exceeds the chains, the tendency is to curl toward hydrocarbon—the polar heads have their largest exposure to water, as occurs for single-chain surfactants, lysolipids, and gangliosides [see **Figure 3*d*** (*top*)]. The positive spontaneous curvature (H_0) is expressed through the assembly of small micelles or the normal hexagonal (H_I) phase (or wormlike micelles), with the head groups outside and the chains inside the aggregate (oil-in-water dispersion). But amphiphiles with smaller head groups or larger chains, as in the case of phospholipids with two acyl groups, require less exposure to water. They favor a more contracted membrane surface, with a smaller interfacial area per lipid. If the optimal head-group separation matches the chains, there is only a small tendency of a monolayer to bend, as in the case of PCs. Hence, the spontaneous curvature H_0 is approximately zero, and the planar lipid bilayer is formed, as in the standard FMM [see **Figure 3*d*** (*middle*)]. Last of all, the lipids with small head groups are even less hydrated, yielding further condensation of the membrane surface. The optimal polar head-group separation is now less than the chains, and the lipid monolayer tends to curl toward water, as in the case of unsaturated and polyunsaturated PEs with a negative spontaneous curvature. The reverse hexagonal (H_{II}) (or cubic) phases are stabilized [see **Figure 3*d*** (*bottom*)], with the head groups inside and the chains outside the lipid microstructure (water-in-oil dispersion).

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Language of Shape for Membrane Lipid–Protein Interactions

In particular, the above curvatures are not implicit; more precisely, they correspond to bending of a neutral (pivotal) plane running beneath the membrane aqueous interface, where the lateral area remains constant (130). Here, what is most arresting is not so much the topology of these fascinating nanostructures, but instead the monolayer curvature, which differs from the canonical planar bilayer geometry. Lipids that adopt nonlamellar phases, such as the H_{II} phase, have a negative spontaneous curvature H_0 (toward water). When they occur in a planar bilayer, there is a mismatch of the geometric mean curvature H (which is zero) from the spontaneous curvature H_0. The two monolayers are held together by the hydrophobic effect and packing forces. Curvature mismatch entails the tendency of an individual monolayer of the bilayer to achieve its natural curvature, which is frustrated by the chain-packing interactions with the other monolayer of a proteolipid membrane (**Figure 4*b***). Despite that a bilayer is flat on average, the two monolayers can still have an intrinsic tendency to curl (19, 60, 140) that can affect the energetics of the lipid–protein interactions.

Strong and Weak Proteolipid Couplings

Now let us return to the influences of proteins on the lipid polymorphism. Compared to the order–disorder phase transitions, the situation is different for the lamellar to nonlamellar phase transitions of the lipids. For unsaturated lipids, at higher temperature, a liquid-crystalline state transition yields the reverse hexagonal (H_{II}) or cubic phases. The latter are familiar as LCPs used for crystallization of membrane proteins (77). A competition of the curvature elastic energy of the monolayer with the chain stretching energy (60, 130) governs the shapes of the lipid nanostructures. Proteins generally tend to stabilize the lamellar versus the nonlamellar phases in the case of phospholipids, meaning there is a strong coupling to the shape deformation in the liquid-crystalline state (43, 111). The coupling does not involve just the liquid-like chains, but rather the curvature elastic stress of the liquid-crystalline membrane (19, 163). Two-way coupling can occur, in which the proteins affect the lipids and the lipids affect the proteins (104). By introducing the Helfrich curvature free energy (see below), one has a ready formulation of the lipid influences in terms of a balance with the hydrophobic matching to the proteolipid interface. Quantitative calculations show that very large values of the curvature free energy can be achieved, giving a potent source of work for modulating protein conformation changes (20).

CURVATURE FORCES IN SOFT BIOMEMBRANES

The role of curvature elastic stress in governing functional membrane protein conformational changes was first proposed on the basis of experimental studies of rhodopsin (163). Conformational changes of rhodopsin can be monitored directly by using optical methods (ultraviolet-visible and Fourier transform infrared spectroscopy) to establish a connection between the lipid properties and protein functional states (22). The main conclusions are that the membrane lipids can significantly affect membrane proteins and that the lipid influences can be chemically nonspecific, indicating that biophysical or material properties are involved. Additional chemically specific interactions may also come into play, as insightfully discussed by Lee (92–94). Experimental tests of the underlying biophysical concepts have been carried out (15, 152, 153, 156), supporting the validity of a collective flexible surface model.

Curvature Stress Field of Proteolipid Membranes

First, let us consider the known unknowns that are due to the lateral pressure profile. We can formulate its influence (31, 76, 103, 158) on membrane protein conformation, either in terms of

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area deformation or by introducing a curvature force field. Establishing how the lateral pressure profile affects lipid–protein interactions is equivalent to a development in terms of the curvature free energy (19, 103). Accordingly, the lateral pressure corresponds to the elastic deformation of the monolayer leaflets of the bilayer, yielding frustration of the spontaneous curvature (H_0) (19). Next, we consider the unknown unknowns that amount to the curvature elastic force field throughout the membrane. In such cases, one needs to conduct membrane simulations (29, 125). Even so, we can boldly cast out the molecular details and instead adopt a simpler continuum treatment (87, 108). The Helfrich approach of using differential geometry and curvature elasticity to explain membrane shape has afforded insights at the cellular membrane level (9, 29, 65, 129, 165), and here, a continuum view is appropriate, because the membrane thickness is much less than its area. But it takes a leap of imagination to extend the approach to the mesoscopic regime, intermediate between the molecular size and the macroscopic dimensions. In fact, that was done for proteolipid membranes (19), thus enabling previously confusing results to be simply explained (20).

Frustration: the increase of the curvature free energy of a lipid monolayer that underlies the stress field of the bilayer

Gaussian curvature: the product of the two principal curvatures that govern the topology of a regular surface

Bending modulus: the energy required to deform the curvature of a flexible surface

Gauss-Bonnet theorem: the mathematical relation between the local geometric properties of a surface (total curvature) and its global topology (Euler characteristic)

The Helfrich Curvature Free Energy

What is important at this point is that the curvature free energy gives a direct link to experimental observables. In keeping with the FSM (**Figure 4*b***), the elastic contribution is viewed in terms of the curvature frustration at the proteolipid boundary, which balances the hydrophobic mismatch between the protein and the lipids. The curvature free energy is due to bending a membrane film (monolayer) and is frustrated by solvation of the hydrophobic protein surface, whereby the two terms in the free energy cannot be simultaneously minimized. Following Landau & Lifschitz (91) or Helfrich (64) then allows a formulation in terms of the mean curvature and the Gaussian curvature of the membrane film, together with the corresponding elastic moduli. Deformation of the mean curvature away from the spontaneous (mean) curvature, together with the Gaussian (saddle) curvature, accounts for the energy of bending a lipid leaflet. The monolayer curvature free energy is balanced (frustrated) by the chain packing energy, owing to stretching of the lipid acyl groups. Nanostructures and self-assembly of the membrane lipids (60) are thus explained, as well as the energetics of the lipid–protein interactions (19, 21).

Considering the curvature elasticity of thin films of amphiphiles offers further insight (9, 10, 17, 29). One can formulate the stresses explicitly in terms of the two principal curvatures (**Figure 4*a***) (64), together with the bending modulus for the mean curvature and the modulus of Gaussian curvature (saddle splay). In differential geometry, any surface is defined by the curvatures due to the intersection of a plane containing the normal vector at any given point. The normal curvature varies with the orientation of the intersecting plane within the vector field along the surface. The maximum and minimum values are defined as the two principal curvatures c_1 and c_2 of the surface at point P, where at regular points, the two principal directions are orthogonal (**Figure 4*a***). The principal curvatures can then be combined into the mean curvature $H = (1/2)(c_1 + c_2)$ with dimensions of inverse length, and the Gaussian curvature $K = c_1 c_2$ with dimensions of inverse area. Notably, the Gaussian curvature is invariant to bending; so if we bend (deform) one simply connected surface to another without stretching, it remains the same at every point. The Gauss-Bonnet theorem describes how the local geometry of a surface is related to its global topology, giving the various lipid nanostructures. The integral of the Gaussian curvature over the surface (known as the integral curvature) is linked to its topology by the Euler characteristic, which describes the connectedness of various surfaces. In turn, the Euler characteristic is related to the genus (the number of holes or handles) of a surface. Hence,

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all surfaces with the same genus have the same integral curvature and can be continuously transformed (bent) from one shape to another.

It follows that the elastic energy due to curvature deformation of a surface can be formulated by the mean curvature and the Gaussian curvature, as defined above, together with the corresponding moduli (64). According to Landau & Lifshitz (91), the elastic energy is treated in a harmonic (Hookean) approximation as a quadratic function of the derivatives of the normal vector along the two principal directions (i.e., the principal curvatures). By variable substitution, the curvature free energy for deforming a soft (flexible) surface thus reads (64),

$$G_c = \oint_S \mathrm{d}g_c \mathrm{d}A = \oint_S \left[\tfrac{1}{2}\kappa \left(\frac{1}{r_1} + \frac{1}{r_2} \right)^2 + \frac{\bar{\kappa}}{r_1 r_2} \right] \mathrm{d}A, \qquad 1.$$

where $r_1 = 1/c_1$ and $r_2 = 1/c_2$ are the two principal radii of curvature for a given geometrical (topological) shape (**Figure 4*a***). Here, $H = (1/2)\ (1/r_1 + 1/r_2)$ is the mean curvature, H_0 is the spontaneous mean curvature (where for a fixed topology, the bending energy is minimized), and $K = 1/r_1 r_2$ is the Gaussian curvature. In addition, the mean curvature bending modulus is κ, and $\bar{\kappa}$ is the modulus of Gaussian curvature. The above expression corresponds to deforming an element of area $\mathrm{d}A$ to a shape described by (r_1, r_2), where integrating over the surface (S) gives the total energy (e.g., due to the lipid/water interface). Equation 1 allows for a semiempirical formulation of the energy of deforming a monolayer film, where Marsh (105) has tabulated experimental values for the elastic constants.

Using Equation 1 we can generally expand the free energy density in the two principal curvatures, $c_1 = 1/r_1$ and $c_2 = 1/r_2$, where other linear combinations are of course possible. By expressing the free energy in terms of the mean curvature $(1/2)\ (1/r_1 + 1/r_2)$, one can separate out a Gaussian curvature term $1/r_1 r_2$, which obeys the Gauss-Bonnet theorem (72) and leads to considerable simplification. The resulting free energy density per unit area depends explicitly on the curvatures and the corresponding elastic moduli. For one monolayer of a lipid bilayer, the following remarkably simple form gives the free energy density (64):

$$g_c = \kappa (H - H_0)^2 + \bar{\kappa} K. \qquad 2.$$

Upon integrating over the surface, the curvature free energy is then obtained because of the displacement of the mean curvature from the spontaneous curvature, together with the Gaussian curvature. Here we note that if the curvature free energy is not at a minimum, then the system is referred to as frustrated (4).

Curvature Versus Hydrophobic Forces

For membrane systems with a finite bending free energy, there must be additional free energy terms (i.e., in the chemical potential) that counterbalance the curvature free energy (e.g., due to solvating the nonpolar protein surface by the lipids) (8, 13). Let us now consider two states of a membrane protein (designated 1 and 2) in a mixture of two lipid types (denoted i and j). The hydrophobic thickness corresponds to the intramembranous protein surface, where A_1 and A_2 are the areas in the two states, and H_1 and H_2 are the lipid mean curvatures (**Figure 4*b***). The spontaneous (intrinsic) monolayer curvature is assumed to be related to the polymorphism of lamellar or nonlamellar-forming lipids. It is given by a sum rule: $H_0 = H_0^i X_i + H_0^j X_j$, where X_i and X_j are the mole fractions of the two lipids (e.g., lamellar $\equiv i$, nonlamellar $\equiv j$) in the mixture. For simplicity, we assume that lipid i favors the planar bilayer (with zero H_0), whereas lipid j favors the nonlamellar (reverse hexagonal) state (with $H_0 < 0$), leading to $H_0 \approx H_0^j X_j$ for the spontaneous

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curvature. The free energy change per protein molecule includes the frustration of the curvature free energy together with the proteolipid lipid surface energy, which to first order reads:

$$\Delta G^{o}/N_{A} = -\kappa(A_{L}H_{2}H_{0}^{j})N_{L}X_{j} + \gamma\Delta A, \quad 3.$$

$$= -\kappa(A_{L}H_{2}N_{L})H_{0} + \gamma\Delta A, \quad 4.$$

where $H_1 \approx 0$ implicitly and to linear order $H_2 \ll 1$. Note that in Equations 3 and 4, H_0^j is the spontaneous curvature because of the nonlamellar lipids, and $\Delta A \equiv A_2 - A_1$, where γ is the proteolipid surface tension, which is assumed the same for both lipid types. (The $\gamma\Delta A$ term corresponds to the proteolipid surface energy; it is a measure of the repulsive part of the nonpolar solvation energy.) In addition, A_L denotes the cross-sectional area per lipid (128), N_L is the number of lipids per protein, and all other symbols have their usual meanings. The Gaussian (saddle splay) curvature is neglected, and contributions from the protein free energy are not included. One can assume the value of H_0 is approximately given by the radius of curvature (R_W) of the reverse hexagonal phase (H_{II}) nanotubes of the lipids under conditions of dual solvent stress (130).

In Equations 3 and 4, the first term is negative and so it yields a favorable driving force to produce the conformational change. Release of the curvature frustration balances the second positive term, as a result of the work of increasing the protein hydrophobic surface area. The FSM predicts the free energy change is linear in the mole fraction (X_j) of the nonlamellar-forming lipid in the mixture (Equation 3). Furthermore, the free energy change depends on the lipid to protein molar ratio (N_L), as strikingly confirmed for rhodopsin (15). The equilibrium constant for the protein transition is related to the spontaneous curvature and proteolipid interfacial tension (Equation 4), where $K = \exp(-\Delta G^{o}/RT)$. Notably, the FSM is a simple robust theory with a minimum of mathematical detail for explaining the lipid–protein interactions in terms of the stress field of the lipid bilayer. It can be elaborated upon by more detailed elastic theories without altering the basic conceptual pillars.

Power in Curvature

The free energy density given by Equation 2 clearly illustrates how curvature forces can affect the membrane deformation. Experimentally, the most significant contribution comes from the bending modulus (rigidity) κ and the spontaneous H_0 curvature. For a lipid bilayer, let us assume representative experimental values of $2\kappa = 4 \times 10^{-19}$ J (105), with a mean monolayer curvature of 1/(40 Å) (51, 62), an area per lipid of 70 Å^2 (128) and 100 lipids/protein molecule. Applying Equation 2 to a planar membrane then gives a curvature free energy of ≈500 kJ/mol protein, a rather large value. By comparison, only approximately 10 kJ/mol protein are needed to effectively shift a conformational equilibrium from reactants to products. At this point, we can conclude that ample free energy is stored within the stress field of the bilayer to drive protein conformational changes in membranes (19).

THE FLEXIBLE SURFACE MODEL

As put forth originally, the flexible surface model (FSM) is an extension or alternative to the classical fluid-mosaic model (FMM) (13, 15, 19–21, 52). A material or colloidal science picture is introduced for the membrane lipid–protein interactions at the mesoscopic length scale, falling in between the macroscopic membrane dimensions and the size of the molecules. Material or biophysical properties due to membrane elasticity and proteolipid solvation (hydrophobic matching) come into play, together with molecularly specific lipid–protein interactions (e.g., due to the lipid head

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FLEXIBLE SURFACE MODEL

The flexible surface model (FSM) extends the fluid-mosaic model of membranes to include proteolipid coupling with the curvature elastic stress field of the lipid bilayer. It accounts for the influences of membrane lipids on protein functional states through nonspecific biophysical or material properties of the lipids. Changes in protein shape can affect hydrophobic matching of the bilayer thickness to the intramembranous protein surface, accompanied by changes in membrane curvature. An energy penalty results from altering the monolayer (leaflet) curvature of the bilayer from the spontaneous (intrinsic) curvature of the lipid film. Release of the curvature frustration coupled to a protein shape change can drive the appearance of new functional states, triggered by environmental cues or changes in allosteric coupling to ligands (agonists or antagonists). Neglect of molecular details is both the strength and weakness of the continuum picture. The FSM explains previously confusing results for various membrane proteins, including GPCRs and ion channels, in a unified manner.

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groups). As noted above, the standard FMM considers only the lateral (in-plane) forces acting on the embedded proteins. Due to the weak proteolipid couplings, the lipid bilayer acts as a mere solvent for the membrane proteins, facilitating their rotational and lateral diffusion. Coupling to longitudinal (out-of-plane) forces is not considered, which are potentially much stronger (19). On the contrary, the newer FSM describes the strong out-of-plane couplings from the curvature forces. Recent attention stems from proteins or peptides that sense or induce curvature, including GPCRs, mechanosensitive channels, and antimicrobial peptides. A lack of molecular specifics is both the strength and weakness of the simple continuum theory. (See the sidebar entitled Flexible Surface Model.)

Two Sides of Membrane Lipids

Because it is a minimal theory, the FSM readily explains how the curvature elastic energy controls membrane protein conformational changes (6, 40, 87, 108). There are two interfaces of interest: namely, the lipid/water interface and the protein/lipid interface (43, 53, 163). Lipid interactions with water (hydrophobic effect) and with the nonpolar protein surface (solvation) yield small differences in large opposing forces that can affect lipid–protein interactions (15, 19, 53). **Figure 4*b*** illustrates how chemically nonspecific bilayer properties can involve bending of the proteolipid membrane (19, 52). Here, the spontaneous monolayer curvature controls the protein conformational changes, together with the elastic moduli. Deformation of the mean curvature H of the flexible surface (neutral plane) away from the spontaneous curvature H_0 leads to curvature strain. The matching curvature stress entails the bending modulus κ, yielding the work (free energy) of the elastic membrane deformation. Influences of nonlamellar-forming lipids (60, 140) on membrane protein function (7, 20, 39, 43, 48, 74, 75, 121, 134, 163) clearly point to an influence of curvature elastic forces (53) on the lipid–protein interactions (19, 71, 127, 163). A key prediction of the FSM is that the nonlamellar-forming membrane lipids can modulate the energetics of integral membrane proteins, as first shown experimentally for rhodopsin (19, 163) and subsequently for mechanosensitive ion channels (127, 129).

Bending and Stretching in Membrane Deformation

This line of thinking takes us to the new biomembrane model, called the flexible surface model. [Actually the FSM has been under continual development since the 1980s (19, 52, 163).]

Lipid–protein interactions are explained by a language of shape (19, 72), where rhodopsin gives a specific example. A membrane protein interacts with the curvature stress field of the membrane lipid bilayer, which is modeled as a continuous liquid-crystalline medium. The FSM explains functional influences of membrane lipid–protein interactions by elastic deformation of the lipid film (15, 19, 125). If a change in protein shape occurs (36, 83, 149, 154), then the free energy of the entire system (comprising protein, lipids, and water) is affected by the membrane bilayer (13). At this level, the contribution to the free energy change is described by two quantities: the bending modulus κ and the monolayer H_0 spontaneous curvature. Deformation of the monolayer curvature H away from its spontaneous (intrinsic) curvature H_0 gives a free energy change that is balanced (frustrated) by the proteolipid solvation. The curvature elastic energy in a given protein state is approximated by $\kappa|H - H_0|^2$, where the Gaussian (saddle splay) curvature (64) is neglected in a first approximation; see Equation 2.

In consequence, matching of the spontaneous (intrinsic) curvature of a lipid monolayer to the proteolipid boundary curvature gives a lipid-mediated force that controls the work of membrane protein conformational changes (**Figure 4*b***). Notably, the spontaneous curvature (H_0) is due to bending of a neutral (pivotal) plane, where the area is constant. As discussed above, it can be toward hydrocarbon ($H_0 > 0$), planar ($H_0 = 0$), or toward water ($H_0 < 0$). The local curvature stress is proportional to $\kappa|H - H_0|$, where κ is the curvature elastic modulus (analogous to Hooke's law). Hydrophobic coupling involves local compression or expansion of the bilayer ($d_P \equiv$ thickness adjacent to proteolipid interface; $d_L \equiv$ thickness of unperturbed bilayer). The frustration entails bending of the neutral plane together with expansion or compression, which competes with solvation of the protein surface. Thus, curvature deformation (remodeling) occurs together with hydrophobic matching at the proteolipid boundary (52, 53).

Relief of the curvature frustration of the individual lipid monolayers can then offset the energetic cost of stretching (or compressing) the lipid acyl chains, because of the hydrophobic mismatch. The balance can be tipped by changes in the membrane interaction that is due to the elastic curvature force field (19). Let us consider a membrane protein with two conformational states in equilibrium (MI and MII for rhodopsin), which involve a protrusion from the lipid bilayer (137). If the second state has a larger hydrophobic interface than the first, then work is needed for the conformational change. Release of the curvature frustration due to lipids having a negative spontaneous curvature (like PE) (**Figure 4*b***) can occur, which stabilizes the protruded (thicker) state of the protein. For example, in the case of rhodopsin, lipids with approximately zero spontaneous curvature H_0 back shift the equilibrium toward the inactive MI state of rhodopsin, whereas lipids with negative H_0 forward shift it toward the active MII state (53, 163). Experimental support comes from plasmon-waveguide resonance spectroscopy (137), which shows a protrusion of rhodopsin from the membrane upon light activation, together with more recent X-ray structures of active rhodopsin (36). Lengthening of TM helix H5 occurs together with a change in the tilt of helix H6 (2, 36), yielding a striking molecular confirmation of the FSM predictions (19).

Detailed theoretical analyses give added value (40, 69, 87, 108, 115, 122, 125) to the basic picture through exact mathematical formulations. According to the FSM, long-range curvature forces due to the lipids can play an important role in controlling protein lateral distributions (e.g., oligomerization) in membranes. Membrane curvature-driven protein–protein interactions have been previously discussed for GPCRs (15) within the context of general bilayer-mediated lipid influences. As proposed by Botelho et al. (15), membrane proteins can interact through the curvature stress field of the bilayer. Simply put, they sense the interactions mediated through the membrane itself, as in the case of rhodopsin (19). A similar role of membrane curvature forces in stabilizing raft-like lipid clusters incorporating cholesterol can occur. Greater local bilayer thickness due to cholesterol [in conjunction with an umbrella effect (26)] can be compensated by a change in curvature of the surrounding lipid membrane (see **Figure 4*b***).

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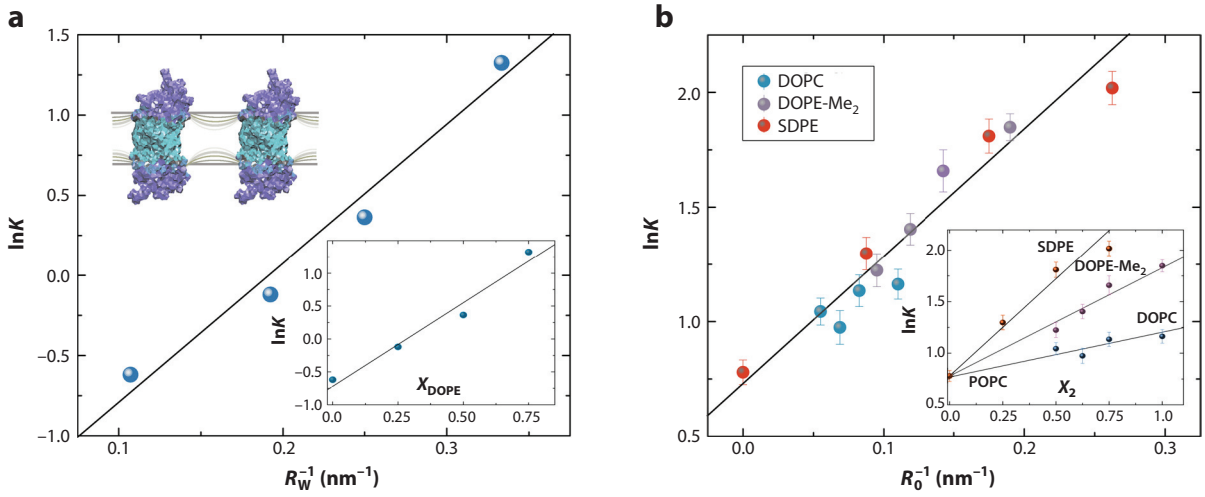

Figure 5

Curvature free energy explains influences of phosphatidylethanolamine (PE) lipids on rhodopsin light activation in membranes. (*a*) The MI–MII equilibrium constant (ln*K*) plotted as a function of spontaneous curvature for a series of PE recombinants containing rhodopsin at $T = 28°C$. For the DOPE/DOPC mixtures, the value of H_0 is given by the inverse water-core radius (R_W) of the H_{II}-phase cylinders ($H_0 = 1/R_W$) (Equation 4). (*Inset*) Corresponding plots versus mole fraction (X_{DOPE}) of DOPE lipids. (*b*) Plot of MI–MII equilibrium constant (ln*K*) against the spontaneous curvature for membranes containing rhodopsin with a series of head-group methylated PEs and polyunsaturated SDPE lipids at $T = 37°C$. The value of H_0 is given by the inverse radius of curvature ($H_0 \equiv 1/R_0$). (*Inset*) Plots of data versus mole fraction of PE lipids. The results collapse to a universal plot versus the spontaneous curvature of the PE-containing lipid mixtures. Rhodopsin data are from References 13 and 153, and lipid values are from References 51, 62, and 79.

Curvature Free Energy Landscape and Protein Functional States

A key predication of Equation 3 is that the lipid to protein ratio influences the conformational energetics, as first discovered for rhodopsin (14) and as confirmed by three independent sets of measurements (15, 123, 152). One possibility is to introduce a persistence length for hydrophobic mismatch of the lipids to the proteolipid boundary; yet the lipid spontaneous curvature is already experimentally measurable (51, 62, 105). Next, we consider the second FSM prediction: According to Equations 3 and 4, the free energy change is linear in the mole fraction (X_j) of nonlamellar-forming lipids in the mixture (13, 19, 52). For rhodopsin, **Figure 5** summarizes the effects of unsaturated PEs in mixtures with PC (13, 153). Recall that unsaturated PEs have negative H_0 (curvature toward water), whereas $H_0 \approx 0$ for PCs (small tendency to curve) (157). In **Figure 5*a***, the equilibrium constant (ln*K*) for the light-activated MI–MII equilibrium of rhodopsin is plotted versus the effective spontaneous curvature for a series of DOPE/DOPC recombinant membranes. The value of H_0 is given by the inverse water-core radius (R_W) of the H_{II}-phase cylinders under dual solvent stress (130). The approximately linear dependence is consistent with the predictions of the FSM (13) (Equation 4), where the free energy change is given by $\Delta G^o = -RT\ln K$. Next, we consider recombinants of rhodopsin with a series of head-group methylated PEs and polyunsaturated SDPE lipids (153) (**Figure 5*b***). Here again, plotting ln*K* against the inverse radius of spontaneous curvature ($R_0 \equiv 1/H_0$) yields an approximately linear relation. For all the PE-containing membranes, a universal linear relation is discovered. The positive linear slope is consistent with a negative curvature (H_2) (toward water) at the proteolipid boundary, in striking agreement with the FSM theory (13, 19, 52). Alternatively, the influences of PE could be due

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to its hydrogen-bonding capacity, its lower hydration, or the small size of the head group versus PC. Yet the effects can all be folded together with the chain pressure into the spontaneous curvature (H_0), as the key material parameter that explains the lipid influences on rhodopsin activation. Closely related formulations have been suggested for mechanosensitive channels (127, 129), where coupling of the protein shape to mechanical curvature forces might seem more obvious.

BAR domain: proteins of the Bin/amphiphysin/Rvs superfamily that are involved in generating membrane curvature and cellular membrane remodeling

MEMBRANE SHAPE TRANSITIONS AND BILAYER REMODELING

As discussed above, curvature deformation is a unifying concept that operates across various length scales, tying together diverse membrane phenomena. In cellular biology, interactions between the lipids and proteins yield a range of possible morphologies, in which membranes can alter their shapes in highly arresting ways (5, 50, 110). Curvature-inducing or curvature-sensing proteins bound to lipid bilayers yield dramatic structural changes in cell membranes, allowing them to participate in movement, cell division, and vesicle trafficking. The various ways proteins interact with lipids to generate curvature remain under active investigation, particularly at the early stages of lipid membrane remodeling (5, 10, 164, 165).

Curvature-Inducing and Curvature-Sensing Proteins

Undoubtedly, the subject of cellular membrane shape transformations has gathered much interest lately, as a result of the role of curvature-sensing proteins and curvature-inducing proteins (10, 50, 110, 146, 147). The BAR (Bin/amphiphysin/Rvs) domain superfamily includes amphiphysin and endophilin, which are involved in the induction of membrane curvature (50, 110) and may participate in clathrin-mediated endocytosis. Because the size of the membrane is much greater than the thickness, use of a two-dimensional approach in terms of differential geometry is appropriate in this case (129, 165). Membrane shape transitions and bilayer remodeling is a vast subject area, and excellent reviews and descriptions can be found (5, 110, 129, 147, 165). Membrane-shaping proteins of the BAR domain superfamily have important roles in membrane trafficking, organelle biogenesis, cell division, and cell migration.

Membrane Remodeling: The Shape of Things to Come

At the protein level, the insertion of amphipathic helices into the bilayer is intimately involved in generating curvature. Cryo–electron microscopic reconstructions have revealed a high level of structural organization among BAR proteins on membrane tubules (112). In addition, coarse-grain MD models of BAR domains with an N-terminal helix (N-BAR) (146) show that binding of N-BAR domain proteins involves their linear aggregation and the formation of meshes on the membrane surface. Voth and colleagues (146) have further shown that the local stress imposed by the N-BAR proteins results in deep invaginations and endocytotic vesicular deformations, in which proteins assemble at the emerging membrane buds. The growing numbers of diseases that result from BAR protein dysfunction suggest new therapeutic strategies in translational medicine (50). In all of these applications, considering membranes as liquid-crystalline soft matter occupies a special place in cellular membrane biophysics.

CONCLUSIONS AND FUTURE PERSPECTIVES

Predicting the future with its inevitable twists and curves is always a risky business. Among the long-term goals of researchers in biophysics is to identify how membrane lipid properties

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underlie protein functions as a foundation for biomedical or biotechnological applications. Whenever tightly controlled metabolic processes go awry, the biological consequences tend to disease. Knowing how dysfunction and dysregulation occur at the cellular membrane level gives inroads into therapeutic intervention in cardiovascular disease, cancer, and visual diseases. Elastic restructuring of the lipids can be implicated in the conformational changes of membrane proteins involving folding, stability, and cellular shape transitions (3, 19, 21, 77, 129, 156). The new biophysics entails curvature matching of the lipids, together with solvation of the nonpolar proteolipid interface. Accordingly, the spontaneous curvature is the property that describes the polymorphism and energetics of the membrane lipids, as well as biological functions of GPCRs, transporters, and ion channels. Any process that occurs within the stress field of the membrane can be affected by the curvature free energy of the lipid bilayer (19)—for example, viral budding, membrane trafficking and biogenesis, and protein folding (116, 162). Understanding how lipids govern the protein states within the bilayer clearly motivates the new biophysics at the convergence of both structure and function.

SUMMARY POINTS

1. The standard fluid-mosaic model assumes that lipids are weakly coupled to the proteins and does not readily explain their influences on membrane functions.
2. Newer models for lipid–protein interactions include hydrophobic matching, raft-like nanodomains, critical phenomena, and membrane curvature deformation.
3. Hydrophobic matching entails short-range forces, where membrane proteins are solvated by a single shell of boundary lipids in analogy to solvation of globular proteins by water.
4. Significant influences of lipids occur in membranes lacking cholesterol, where raft-like nanodomains or a miscibility critical point are unlikely.
5. Nonlamellar-forming lipids together with long-range forces suggest that proteins are affected by the curvature elasticity of the lipid bilayer.
6. The flexible surface model explains how biophysical properties of the lipids govern the photochemical function of rhodopsin as a prototype for other integral membrane proteins.

DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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LITERATURE CITED

1. Ackerman DG, Feigenson GW. 2015. Multiscale modeling of four-component lipid mixtures: domain composition, size, alignment, and properties of the phase interface. *J. Phys. Chem. B* 119:4240–50
2. Altenbach C, Kusnetzow AK, Ernst OP, Hofmann KP, Hubbell WL. 2008. High-resolution distance mapping in rhodopsin reveals the pattern of helix movement due to activation. *PNAS* 105:7439–44
3. Andersen OS, Koeppe RE II. 2007. Bilayer thickness and membrane protein function: an energetic perspective. *Annu. Rev. Biophys. Biomolec. Struct.* 36:107–30
4. Anderson D, Wennerström H, Olsson U. 1989. Isotropic bicontinuous solutions in surfactant-solvent systems: the L_3 phase. *J. Phys. Chem.* 93:4243–53
5. Antonny B. 2011. Mechanisms of membrane curvature sensing. *Annu. Rev. Biochem.* 80:101–23
6. Aranda-Espinoza H, Berman A, Dan P, Pincus P, Safran S. 1996. Interaction between inclusions embedded in membranes. *Biophys. J.* 71:648–56
7. Attard GS, Templer RH, Smith WS, Hunt AN, Jackowski S. 2000. Modulation of CTP:phosphocholine cytidylyltransferase by membrane curvature elastic stress. *PNAS* 97:9032–36
8. Baldwin PA, Hubbell WL. 1985. Effects of lipid environment on the light-induced conformational changes of rhodopsin. 2. Roles of lipid chain length, unsaturation, and phase state. *Biochemistry* 24:2633–39
9. Bassereau P, Sorre B, Lévy A. 2014. Bending lipid membranes: experiments after W. Helfrich's model. *Adv. Colloid Interface Sci.* 208:47–57
10. Baumgart T, Capraro BR, Zhu C, Das SL. 2011. Thermodynamics and mechanics of membrane curvature generation and sensing by proteins and lipids. *Annu. Rev. Phys. Chem.* 62:483–506
11. Bloom M, Evans E, Mouritsen OG. 1991. Physical properties of the fluid lipid-bilayer component of cell membranes: a perspective. *Q. Rev. Biophys.* 24:293–397
12. Bogdanov M, Heacock P, Guan Z, Dowhan W. 2010. Plasticity of lipid-protein interactions in the function and topogenesis of the membrane protein lactose permease from *Escherichia coli*. *PNAS* 107:15057–62
13. Botelho AV, Gibson NJ, Wang Y, Thurmond RL, Brown MF. 2002. Conformational energetics of rhodopsin modulated by nonlamellar forming lipids. *Biochemistry* 41:6354–68
14. Botelho AV, Huber T, Brown MF. 2003. Flexible surface model for lipid-rhodopsin interactions: further analysis. *Biophys. J.* 84:55A
15. Botelho AV, Huber T, Sakmar TP, Brown MF. 2006. Curvature and hydrophobic forces drive oligomerization and modulate activity of rhodopsin in membranes. *Biophys. J.* 91:4464–77
16. Brown DA, London E. 1998. Functions of lipid rafts in biological membranes. *Annu. Rev. Cell Dev. Biol.* 14:111–36
17. Brown FLH. 2008. Elastic modeling of biomembranes and lipid bilayers. *Annu. Rev. Phys. Chem.* 59:685–712
18. Brown MF. 1982. Theory of spin-lattice relaxation in lipid bilayers and biological membranes. ^{2}H and ^{14}N quadrupolar relaxation. *J. Chem. Phys.* 77:1576–99
19. Brown MF. 1994. Modulation of rhodopsin function by properties of the membrane bilayer. *Chem. Phys. Lipids* 73:159–80
20. Brown MF. 1997. Influence of non-lamellar forming lipids on rhodopsin. In *Current Topics in Membranes*, ed. RM Epand, pp. 285–356. San Diego: Academic
21. Brown MF. 2012. Curvature forces in membrane lipid–protein interactions. *Biochemistry* 51:9782–95
22. Brown MF. 2012. UV–visible and infrared methods for investigating lipid–rhodopsin membrane interactions. In *Membrane Protein Structure and Dynamics: Methods and Protocols (Methods in Molecular Biology)*, ed. N Vaidehi, J Klein-Seetharaman, pp. 127–53. New York: Springer Science+Business Media, LLC
23. Brown MF, Miljanich GP, Dratz EA. 1977. Proton spin-lattice relaxation of retinal rod outer segment membranes and liposomes of extracted phospholipids. *PNAS* 74:1978–82
24. Brown MF, Ribeiro AA, Williams GD. 1983. New view of lipid bilayer dynamics from ^{2}H and ^{13}C NMR relaxation time measurements. *PNAS* 80:4325–29
25. Brown MF, Seelig J. 1977. Ion-induced changes in head group conformation of lecithin bilayers. *Nature* 269:721–23

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26. Brown MF, Seelig J. 1978. Influence of cholesterol on the polar region of phosphatidylcholine and phosphatidylethanolamine bilayers. *Biochemistry* 17:381–84
27. Brown MF, Seelig J, Häberlen U. 1979. Structural dynamics in phospholipid bilayers from deuterium spin-lattice relaxation time measurements. *J. Chem. Phys.* 70:5045–53
28. Brown MF, Thurmond RL, Dodd SW, Otten D, Beyer K. 2002. Elastic deformation of membrane bilayers probed by deuterium NMR relaxation. *J. Am. Chem. Soc.* 124:8471–84
29. Campelo F, Arnarez C, Marrink SJ, Kozlov MM. 2014. Helfrich model of membrane bending: from Gibbs theory of liquid interfaces to membranes as thick anisotropic elastic layers. *Adv. Colloid Interface Sci.* 208:25–33
30. Cantor RS. 1997. Lateral pressures in cell membranes: a mechanism for modulation of protein function. *J. Phys. Chem. B* 101:1723–25
31. Cantor RS. 1999. The influence of membrane lateral pressures on simple geometric models of protein conformational equilibria. *Chem. Phys. Lipids* 101:45–56
32. Cantor RS. 1999. Lipid composition and the lateral pressure profile in bilayers. *Biophys. J.* 76:2625–39
33. Chaptal V, Kwon S, Sawaya MR, Guan L, Kaback HR, Abramson J. 2011. Crystal structure of lactose permease in complex with an affinity inactivator yields unique insight into sugar recognition. *PNAS* 108:9361–66
34. Chawla U, Jiang Y, Zheng W, Kuang L, Perera SMDC, et al. 2016. A usual G-protein-coupled receptor in unusual membranes. *Angew. Chem. Int. Ed.* 128:598–602
35. Chawla U, Perera SMDC, Struts AV, Pitman MC, Brown MF. 2016. Hydration mediated G-protein-coupled receptor activation. *Biophys. J.* 110:83a
36. Choe H-W, Kim YJ, Park JH, Morizumi T, Pai EF, et al. 2011. Crystal structure of metarhodopsin II. *Nature* 471:651–55
37. Cronan JE. 2003. Bacterial membrane lipids: Where do we stand? *Annu. Rev. Microbiol.* 57:203–24
38. Curnow P, Lorch M, Charalambous K, Booth PJ. 2004. The reconstitution and activity of the small multidrug transporter EmrE is modulated by non-bilayer lipid composition. *J. Mol. Biol.* 343:213–22
39. Dahlqvist A, Nordström S, Karlsson OP, Mannock DA, McElhaney RN, Wieslander Å. 1995. Efficient modulation of glucolipid enzyme activities in membranes of *Acholeplasma laidlawii* by the type of lipids in the bilayer matrix. *Biochemistry* 34:13381–89
40. Dan N, Pincus P, Safran SA. 1993. Membrane-induced interactions between inclusions. *Langmuir* 9:2768–71
41. Dawaliby R, Trubbia C, Delporte C, Masureel M, Van Antwerpen P, et al. 2016. Allosteric regulation of G protein–coupled receptor activity by phospholipids. *Nat. Chem. Biol.* 12:35–39
42. Day CA, Kenworthy AK. 2009. Tracking microdomain dynamics in cell membranes. *Biochim. Biophys. Acta* 1788:245–53
43. Deese AJ, Dratz EA, Brown MF. 1981. Retinal rod outer segment lipids form bilayers in the presence and absence of rhodopsin: a ^{31}P NMR study. *FEBS Lett.* 124:93–99
44. Dill KA, Flory PJ. 1980. Interphases of chain molecules: monolayers and lipid bilayer membranes. *PNAS* 77:3115–19
45. Dill KA, Truskett TM, Vlachy V, Hribar-Lee B. 2005. Modeling water, the hydrophobic effect, and ion solvation. *Annu. Rev. Biophys. Biomol. Struct.* 34:173–99
46. Dowhan W. 1997. Molecular basis for membrane phospholipid diversity: Why are there so many lipids? *Annu. Rev. Biochem.* 66:199–232
47. Engelman DM. 2005. Membranes are more mosaic than fluid. *Nature* 438:578–80
48. Epand RM. 1998. Lipid polymorphism and protein–lipid interactions. *Biochim. Biophys. Acta* 1376:353–68
49. Fattal DR, Ben-Shaul A. 1993. A molecular model for lipid-protein interaction in membranes: the role of hydrophobic mismatch. *Biophys. J.* 65:1795–809
50. Frost A, Unger VM, De Camilli P. 2009. The BAR domain superfamily: membrane-molding macromolecules. *Cell* 137:191–96
51. Fuller N, Rand RP. 2001. The influence of lysolipids on the spontaneous curvature and bending elasticity of phospholipid membranes. *Biophys. J.* 81:243–54

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52. Gibson NJ, Brown MF. 1991. Membrane lipid influences on the energetics of the metarhodopsin I and metarhodopsin II conformational states of rhodopsin probed by flash photolysis. *Photochem. Photobiol.* 54:985–92
53. Gibson NJ, Brown MF. 1993. Lipid headgroup and acyl chain composition modulate the MI–MII equilibrium of rhodopsin in recombinant membranes. *Biochemistry* 32:2438–54
54. Gonen T, Cheng Y, Sliz P, Hiroaki Y, Fujiyoshi Y, et al. 2005. Lipid-protein interactions in double-layered two-dimensional AQPO crystals. *Nature* 438:633–38
55. Goñi F, Alonso A, Bagatolli LA, Brown RE, Marsh D, et al. 2008. Phase diagrams of lipid mixtures relevant to the study of membrane rafts. *Biochim. Biophys. Acta* 1781:665–84
56. Granier S, Kobilka B. 2012. A new era of GPCR structural and chemical biology. *Nat. Chem. Biol.* 8:670–73
57. Griffith OH, Jost PC. 1976. Lipid spin labels in biological membranes. In *Spin Labeling: Theory and Applications*, ed. LJ Berliner, pp. 453–523. New York: Academic
58. Grossfield A, Feller SE, Pitman MC. 2006. A role for direct interactions in the modulation of rhodopsin by ω-3 polyunsaturated lipids. *PNAS* 103:4888–93
59. Gruner SM. 1985. Intrinsic curvature hypothesis for biomembrane lipid composition: a role for nonbilayer lipids. *PNAS* 82:3665–69
60. Gruner SM. 1989. Stability of lyotropic phases with curved interfaces. *J. Phys. Chem.* 93:7562–70
61. Gullingsrud J, Schulten K. 2004. Lipid bilayer pressure profiles and mechanosensitive channel gating. *Biophys. J.* 86:3496–509
62. Hamai C, Yang T, Kataoka S, Cremer PS, Musser SM. 2006. Effect of average phospholipid curvature on supported bilayer formation on glass by vesicle fusion. *Biophys. J.* 90:1241–48
63. Harroun TA, Heller WT, Weiss TM, Yang L, Huang HW. 1999. Theoretical analysis of hydrophobic matching and membrane-mediated interactions in lipid bilayers containing gramicidin. *Biophys. J.* 76:3176–85
64. Helfrich W. 1973. Elastic properties of lipid bilayers: theory and possible experiments. *Z. Naturforsch.* 28c:693–703
65. Helfrich W, Servuss R-M. 1984. Undulations, steric interaction and cohesion of fluid membranes. *Nuovo Cim.* 3:137–51
66. Honerkamp-Smith AR, Cicuta P, Collins MD, Veatch SL, den Nijs M, et al. 2008. Line tensions, correlation lengths, and critical exponents in lipid membranes near critical points. *Biophys. J.* 95:236–46
67. Honerkamp-Smith AR, Veatch SL, Keller SL. 2009. An introduction to critical points for biophysicists; observations of compositional heterogeneity in lipid membranes. *Biochim. Biophys. Acta* 1788:53–63
68. Hong H, Tamm LK. 2004. Elastic coupling of integral membrane protein stability to lipid bilayer forces. *PNAS* 101:4065–70
69. Huang HW. 1986. Deformation free energy of bilayer membrane and its effect on gramicidin channel lifetime. *Biophys. J.* 50:1061–70
70. Huber T, Botelho AV, Beyer K, Brown MF. 2004. Membrane model for the GPCR prototype rhodopsin: hydrophobic interface and dynamical structure. *Biophys. J.* 86:2078–100
71. Hui SW. 1997. Curvature stress and biomembrane function. In *Current Topics in Membranes*, ed. RM Epand, pp. 541–63. San Diego: Academic
72. Hyde ST, Andersson S, Larsson K, Blum Z, Landh T, et al. 1997. *The Language of Shape. The Role of Curvature in Condensed Matter: Physics, Chemistry and Biology*. Amsterdam: Elsevier
73. Israelachvili JN. 2011. *Intermolecular and Surface Forces*. San Diego: Academic. 3rd ed.
74. Jensen JW, Schutzbach JS. 1984. Activation of mannosyltransferase II by nonbilayer phospholipids. *Biochemistry* 23:1115–19
75. Jensen JW, Schutzbach JS. 1988. Modulation of dolichyl-phosphomannose synthase activity by changes in the lipid environment of the enzyme. *Biochemistry* 27:6315–20
76. Jensen MØ, Mouritsen OG. 2004. Lipids do influence protein function—the hydrophobic matching hypothesis revisited. *Biochim. Biophys. Acta* 1666:205–26
77. Johner N, Mondal S, Morra G, Caffrey M, Weinstein H, Khelashvili G. 2014. Protein and lipid interactions driving molecular mechanisms of in meso crystallization. *J. Am. Chem. Soc.* 136:3271–84

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78. Katritch V, Cherezov V, Stevens RC. 2012. Diversity and modularity of G protein-coupled receptor structures. *Trends Pharm. Sci.* 33:17–27
79. Keller SL, Bezrukov SM, Gruner SM, Tate MW, Vodyanoy I, Parsegian VA. 1993. Probability of alamethicin conductance states varies with nonlamellar tendency of bilayer phospholipids. *Biophys. J.* 65:23–27
80. Khalili-Araghi F, Gumbart J, Wen P-C, Sotomayor M, Tajkhorshid E, Schulten K. 2009. Molecular dynamics simulations of membrane channels and transporters. *Curr. Opin. Struct. Biol.* 19:128–37
81. Khelashvili G, Blecua Carrillo Albornoz P, Johner N, Mondal S, Caffrey M, Weinstein H. 2012. Why GPCRs behave differently in cubic and lamellar lipidic mesophases. *J. Am. Chem. Soc.* 134:15858–68
82. Killian JA. 1998. Hydrophobic mismatch between proteins and lipids in membranes. *Biochim. Biophys. Acta* 1376:401–16
83. Kimata N, Pope A, Eilers M, Opefi CA, Ziliox M, et al. 2016. Retinal orientation and interactions in rhodopsin reveal a two-stage trigger mechanism for activation. *Nat. Comm.* 7:12683
84. Kinnun JJ, Mallikarjunaiah KJ, Petrache HI, Brown MF. 2015. Elastic deformation and area per lipid of membranes: atomistic view from solid-state deuterium NMR spectroscopy. *Biochim. Biophys. Acta* 1848:246–59
85. Kobayashi M, Struts AV, Fujiwara T, Brown MF, Akutsu H. 2008. Fluid mechanical matching of H^+-ATP synthase subunit *c*-ring with lipid membranes revealed by ^2H solid-state NMR. *Biophys. J.* 94:4339–47
86. Koldsø H, Sansom MSP. 2015. Organization and dynamics of receptor proteins in a plasma membrane. *J. Am. Chem. Soc.* 137:14694–704
87. Kralchevsky PA, Paunov VN, Denkov ND, Nagayama K. 1995. Stresses in lipid membranes and interactions between inclusions. *J. Chem. Soc. Faraday Trans.* 91:3415–32
88. Kusnetzow AK, Altenbach C, Hubbell WL. 2006. Conformational states and dynamics of rhodopsin in micelles and bilayers. *Biochemistry* 45:5538–50
89. Kwon B, Waring AJ, Hong M. 2013. A ^2H solid-state NMR study of lipid clustering by cationic antimicrobial and cell-penetrating peptides in model bacterial membranes. *Biophys. J.* 105:2333–42
90. Laganowsky A, Reading E, Allison TM, Ulmschneider MB, Degiacomi MT, et al. 2014. Membrane proteins bind lipids selectively to modulate their structure and function. *Nature* 510:172–75
91. Landau LD, Lifshitz EM. 1986. *Theory of Elasticity*. Oxford, UK: Pergamon. 3rd ed.
92. Lee AG. 1998. How lipids interact with an intrinsic membrane protein: the case of the calcium pump. *Biochim. Biophys. Acta* 1376:381–90
93. Lee AG. 2004. How lipids affect the activities of integral membrane proteins. *Biochim. Biophys. Acta* 1666:62–87
94. Lee AG. 2011. Biological membranes: the importance of molecular detail. *Trends Biochem. Sci.* 36:493–500
95. Leftin A, Brown MF. 2011. An NMR data base for simulations of membrane dynamics. *Biochim. Biophys. Acta* 1808:818–39
96. Lewis JR, Cafiso DS. 1999. Correlation between the free energy of a channel-forming voltage-gated peptide and the spontaneous curvature of bilayer lipids. *Biochemistry* 38:5932–38
97. Lindblom G, Brentel I, Sjölund M, Wikander G, Wieslander Å. 1986. Phase equilibria of membrane lipids from *Acholeplasma laidlawii*: importance of a single lipid forming nonlamellar phases. *Biochemistry* 25:7502–10
98. Lyman E, Cui HS, Voth GA. 2011. Reconstructing protein remodeled membranes in molecular detail from mesoscopic models. *Phys. Chem. Chem. Phys.* 13:10430–36
99. Machta BB, Veatch SL, Sethna JP. 2012. Critical Casimir forces in cellular membranes. *Phys. Rev. Lett.* 109:138101
100. Mahalingam M, Martínez-Mayorga K, Brown MF, Vogel R. 2008. Two protonation switches control rhodopsin activation in membranes. *PNAS* 105:17795–800
101. Marsh D. 1996. Intrinsic curvature in normal and inverted lipid structures and in membranes. *Biophys. J.* 70:2248–55
102. Marsh D. 1996. Lateral pressure in membranes. *Biochim. Biophys. Acta* 1286:183–223

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103. Marsh D. 2007. Lateral pressure profile, spontaneous curvature frustration, and the incorporation and conformation of proteins in membranes. *Biophys. J.* 93:3884–99
104. Marsh D. 2008. Protein modulation of lipids, and vice-versa, in membranes. *Biochim. Biophys. Acta* 1778:1545–75
105. Marsh D. 2013. *Handbook of Lipid Bilayers*. Boca Raton, FL: CRC Press. 2nd ed.
106. Martinez GV, Dykstra EM, Lope-Piedrafita S, Job C, Brown MF. 2002. NMR elastometry of fluid membranes in the mesoscopic regime. *Phys. Rev. E* 66:050902
107. Marty MT, Hoi KK, Gault J, Robinson CV. 2016. Probing the lipid annular belt by gas-phase dissociation of membrane proteins in nanodiscs. *Angew. Chem. Int. Ed.* 55:550–54
108. May S, Ben-Shaul A. 1999. Molecular theory of lipid-protein interaction and the L_α-H_{II} transition. *Biophys. J.* 76:751–67
109. McConnell HM, Vrljic M. 2003. Liquid-liquid immiscibility in membranes. *Annu. Rev. Biophys. Biomol. Struct.* 32:469–92
110. McMahon HT, Gallop JL. 2005. Membrane curvature and mechanisms of dynamic cell membrane remodelling. *Nature* 438:590–96
111. Miljanich GP, Brown MF, Mabrey-Gaud S, Dratz EA, Sturtevant JM. 1985. Thermotropic behavior of retinal rod membranes and dispersions of extracted phospholipids. *J. Membr. Biol.* 85:79–86
112. Mim C, Cui H, Gawronski-Salerno JA, Frost A, Lyman E, et al. 2012. Structural basis of membrane bending by the N-BAR protein endophilin. *Cell* 149:137–45
113. Mitchell DC, Straume M, Miller JL, Litman BJ. 1990. Modulation of metarhodopsin formation by cholesterol-induced ordering of bilayer lipids. *Biochemistry* 29:9143–49
114. Molugu TR, Brown MF. 2016. Cholesterol-induced suppression of membrane elastic fluctuations at the atomistic level. *Chem. Phys. Lipids* 199:39–51
115. Mondal S, Khelashvili G, Shan J, Andersen OS, Weinstein H. 2011. Quantitative modeling of membrane deformations by multihelical membrane proteins: application to G-protein coupled receptors. *Biophys. J.* 101:2092–101
116. Moon CP, Fleming KG. 2011. Side-chain hydrophobicity scale derived from transmembrane protein folding into lipid bilayers. *PNAS* 108:10174–77
117. Morein S, Andersson A-S, Rilfors L, Lindblom G. 1996. Wild-type *Escherichia coli* cells regulate the membrane lipid composition in a "window" between gel and non-lamellar structures. *J. Biol. Chem.* 271:6801–9
118. Mouritsen OG. 2005. *Life—As a Matter of Fat*. Heidelberg, Ger.: Springer
119. Mouritsen OG, Bloom M. 1993. Models of lipid-protein interactions in membranes. *Annu. Rev. Biophys. Biomol. Struct.* 22:145–71
120. Nagle JF, Tristram-Nagle S. 2000. Structure of lipid bilayers. *Biochim. Biophys. Acta* 1469:159–95
121. Navarro J, Toivio-Kinnucan M, Racker E. 1984. Effect of lipid composition on the calcium/adenosine 5′-triphosphate coupling ratio of the Ca^{2+}-ATPase of sarcoplasmic reticulum. *Biochemistry* 23:130–35
122. Nielsen C, Andersen OS. 2000. Inclusion-induced bilayer deformations: effects of monolayer equilibrium curvature. *Biophys. J.* 79:2583–604
123. Niu S-L, Mitchell DC. 2005. Effect of packing density on rhodopsin stability and function in polyunsaturated membranes. *Biophys. J.* 89:1833–40
124. Olesen C, Picard M, Winther A-ML, Gyrup C, Morth JP, et al. 2007. The structural basis of calcium transport by the calcium pump. *Nature* 450:1036–42
125. Ollila OHS, Risselada HJ, Louhivuori M, Lindahl E, Vattulainen I, Marrink SJ. 2009. 3D pressure field in lipid membranes and membrane-protein complexes. *Phys. Rev. Lett.* 102:078101
126. Park SH, Opella SJ. 2005. Tilt angle of a trans-membrane helix is determined by hydrophobic mismatch. *J. Mol. Biol.* 350:310–18
127. Perozo E, Kloda A, Cortes DM, Martinac B. 2002. Physical principles underlying the transduction of bilayer deformation forces during mechanosensitive channel gating. *Nat. Struct. Biol.* 9:696–703
128. Petrache HI, Dodd SW, Brown MF. 2000. Area per lipid and acyl length distributions in fluid phosphatidylcholines determined by ^{2}H NMR spectroscopy. *Biophys. J.* 79:3172–92
129. Phillips R, Ursell T, Wiggins P, Sens P. 2009. Emerging roles for lipids in shaping membrane-protein function. *Nature* 459:379–85

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130. Rand RP, Fuller NL, Gruner SM, Parsegian VA. 1990. Membrane curvature, lipid segregation, and structural transitions for phospholipids under dual-solvent stress. *Biochemistry* 29:76–87
131. Rietveld A, van Kemenade TJJM, Hak T, Verkleij AJ, de Kruijff B. 1987. The effect of cytochrome *c* oxidase on lipid polymorphism of model membranes containing cardiolipin. *Eur. J. Biochem.* 164:137–40
132. Rietveld AG, Chupin VV, Koorengevel MC, Wienk HLJ, Dowhan W, de Kruijff B. 1994. Regulation of lipid polymorphism is essential for the viability of phosphatidylethanolamine-deficient *Escherichia coli* cells. *J. Biol. Chem.* 269:28670–75
133. Rietveld AG, Killian JA, Dowhan W, de Kruijff B. 1993. Polymorphic regulation of membrane phospholipid composition in *Escherichia coli*. *J. Biol. Chem.* 268:12427–33
134. Rietveld AG, Koorengevel MC, de Kruijff B. 1995. Non-bilayer lipids are required for efficient protein transport across the plasma membrane of *Escherichia coli*. *EMBO J.* 14:5506–13
135. Rosenbaum DM, Zhang C, Lyons JA, Holl R, Aragao D, et al. 2011. Structure and function of an irreversible agonist-β_2 adrenoceptor complex. *Nature* 469:236–40
136. Sachs JN, Engelman DM. 2006. Introduction to the membrane protein reviews: the interplay of structure, dynamics, and environment in membrane protein function. *Annu. Rev. Biochem.* 75:707–12
137. Salamon Z, Brown MF, Tollin G. 1999. Plasmon resonance spectroscopy: probing molecular interactions within membranes. *Trends Biochem. Sci.* 24:213–19
138. Salmon A, Dodd SW, Williams GD, Beach JM, Brown MF. 1987. Configurational statistics of acyl chains in polyunsaturated lipid bilayers from ^{2}H NMR. *J. Am. Chem. Soc.* 109:2600–9
139. Schwartz SD, Schramm VL. 2009. Enzymatic transition states and dynamic motion in barrier crossing. *Nat. Chem. Biol.* 5:552–59
140. Seddon JM. 1990. Structure of the inverted hexagonal (H_{II}) phase, and non-lamellar phase transitions of lipids. *Biochim. Biophys. Acta* 1031:1–69
141. Seelig J, MacDonald PM, Scherer PG. 1987. Phospholipid head groups as sensors of electric charge in membranes. *Biochemistry* 26:7535–41
142. Seelig J, Seelig A. 1980. Lipid conformation in model membranes and biological membranes. *Quart. Rev. Biophys.* 13:19–61
143. Siegel DP, Cherezov V, Greathouse DV, Koeppe II RE, Killian JA, Caffrey M. 2006. Transmembrane peptides stabilize inverted cubic phases in a biphasic length-dependent manner: implications for protein-induced membrane fusion. *Biophys. J.* 90:200–11
144. Simons K, Gerl MJ. 2010. Revitalizing membrane rafts: new tools and insights. *Nat. Rev. Mol. Cell Biol.* 11:688–99
145. Simons K, Vaz WLC. 2004. Model systems, lipid rafts, and cell membranes. *Annu. Rev. Biophys. Biomol. Struct.* 33:269–95
146. Simunovic M, Srivastava A, Voth GA. 2013. Linear aggregation of proteins on the membrane as a prelude to membrane remodeling. *PNAS* 110:20396–401
147. Simunovic M, Voth GA, Callan-Jones A, Bassereau P. 2015. When physics takes over: BAR proteins and membrane curvature. *Trends Cell Bio*. 25:780–92
148. Singer SJ, Nicolson GL. 1972. The fluid mosaic model of the structure of cell membranes. *Science* 175:720–31
149. Smith SO. 2010. Structure and activation of the visual pigment rhodopsin. *Annu. Rev. Biophys.* 39:309–28
150. Sodt AJ, Pastor RW. 2013. Bending free energy from simulation: correspondence of planar and inverse hexagonal lipid phases. *Biophys. J.* 104:2202–11
151. Sodt AJ, Sandar ML, Gawrisch K, Pastor RW, Lyman E. 2014. The molecular structure of the liquid-ordered phase of lipid bilayers. *J. Am. Chem. Soc.* 136:725–32
152. Soubias O, Teague WE Jr., Hines KG, Gawrisch K. 2015. Rhodopsin/lipid hydrophobic matching—rhodopsin oligomerization and function. *Biophys. J.* 108:1125–32
153. Soubias O, Teague WE Jr., Hines KG, Mitchell DC, Gawrisch K. 2010. Contribution of membrane elastic energy to rhodopsin function. *Biophys. J.* 99:817–24
154. Standfuss J, Edwards PC, D'Antona A, Fransen MR, Xie G, et al. 2011. The structural basis of agonist-induced activation in constitutively active rhodopsin. *Nature* 471:656–60
155. Struts AV, Salgado GFJ, Martínez-Mayorga K, Brown MF. 2011. Retinal dynamics underlie its switch from inverse agonist to agonist during rhodopsin activation. *Nat. Struct. Mol. Biol.* 18:392–94

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156. Teague WE Jr., Soubias O, Petrache H, Fuller N, Hines KG, et al. 2013. Elastic properties of polyunsaturated phosphatidylethanolamines influence rhodopsin function. *Faraday Discuss.* 161:383–95
157. Thurmond RL, Lindblom G, Brown MF. 1993. Curvature, order, and dynamics of lipid hexagonal phases studied by deuterium NMR spectroscopy. *Biochemistry* 32:5394–410
158. van den Brink-van der Laan E, Killian JA, de Kruijff B. 2004. Nonbilayer lipids affect peripheral and integral membrane proteins via changes in the lateral pressure profile. *Biochim. Biophys. Acta* 1666:275–88
159. van der Wel PCA, Strandberg E, Killian JA, Koeppe RE II. 2002. Geometry and intrinsic tilt of a tryptophan-anchored transmembrane α-helix determined by ^{2}H NMR. *Biophys. J.* 83:1479–88
160. Veatch SL, Cicuta P, Sengupta P, Honerkamp-Smith A, Holowka D, Barbara BB. 2008. Critical fluctuations in plasma membrane vesicles. *ACS Chem. Biol.* 3:287–93
161. Veatch SL, Soubias O, Keller SL, Gawrisch K. 2007. Critical fluctuations in domain-forming lipid mixtures. *PNAS* 104:17650–55
162. White SH, Wimley WC. 1999. Membrane protein folding and stability: physical principles. *Annu. Rev. Biophys. Biomol. Struct.* 28:319–65
163. Wiedmann TS, Pates RD, Beach JM, Salmon A, Brown MF. 1988. Lipid-protein interactions mediate the photochemical function of rhodopsin. *Biochemistry* 27:6469–74
164. Zimmerberg J, Gawrisch K. 2006. The physical chemistry of biological membranes. *Nat. Chem. Biol.* 2:564–67
165. Zimmerberg J, Kozlov MM. 2006. How proteins produce cellular membrane curvature. *Nat. Rev. Mol. Cell Biol.* 7:9–19

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